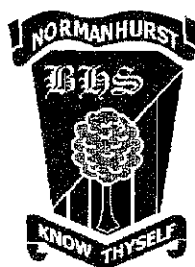


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NESA STUDENT NUMBER



# 2023

## Year 12 Chemistry

### Trial Examination

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#### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen.
- Draw diagrams using pencil.
- Show all relevant working in questions involving calculations.
- NESA approved calculators may be used.

**Total marks: 100**

#### Section 1 – Multiple Choice

**20 marks**

- Attempt Questions 1–20

#### Section 2A – Written Response

**20 marks**

- Attempt Questions 1–4

#### Section 2B – Written Response

**40 marks**

- Attempt Questions 1–8

#### Section 2C – Written Response

**20 marks**

- Attempt Questions 1–4

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NESA STUDENT NUMBER



**2023**

# NBHS Year 12 Chemistry Trial Examination

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample     $2 + 4 =$     (A) 2    (B) 6    (C) 8    (D) 9

A ☐    B ☒    C ☐    D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒    B ☒    C ☐    D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒    B ☒    C ☐    D ☐

*correct*

- |    |                         |                         |                         |                         |
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| 20 | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

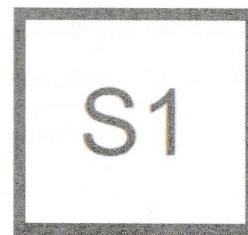
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NESA STUDENT NUMBER

## Section 1

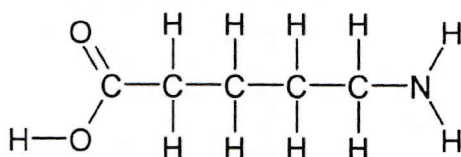
20 marks

- Attempt Questions 1–20.
- Use the multiple-choice answer sheet at the **back of section 2A** to answer the multiple-choice questions for Section 1.



### Question 1 (1 mark)

An organic compound shown below.

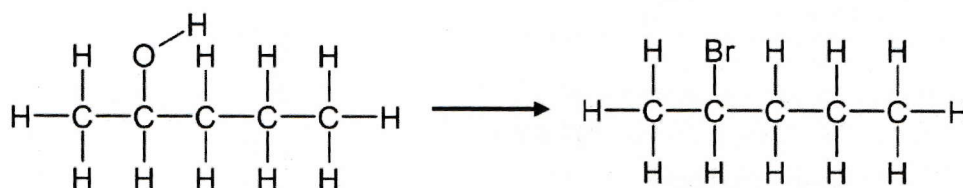


Which are the TWO functional groups present in this molecule?

- (A) Amide and ester.
- (B) Amide and carbonyl.
- (C) Carboxyl and amine.
- (D) Alkene and amine.

### Question 2 (1 mark)

Which of the following conditions would be appropriate to carry out the following synthesis?



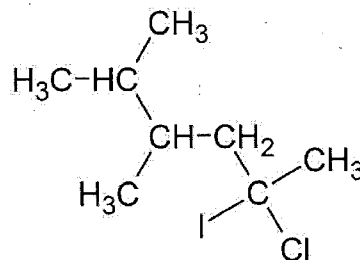
- (A) Br<sub>2</sub> and UV light.
- (B) Br<sub>2</sub> dissolved in water.
- (C) HBr and heat.
- (D) HBr and ZnCl<sub>2</sub> catalyst.

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NESA STUDENT NUMBER

**Question 3 (1 mark)**

The structure of a compound is shown below.



What is the IUPAC name of this compound?

- (A) 2-chloro-2-iodo-4,5-dimethylhexane
- (B) 2-iodo-2-chloro-4,5-dimethylhexane
- (C) 2,3-dimethyl-5-chloro-5-iodohexane
- (D) 5-chloro-5-iodo-2,3-dimethylhexane

**Question 4 (1 mark)**

The heat of combustion of butan-1-ol is  $2676 \text{ kJ mol}^{-1}$ .

What is the value of the heat of combustion in  $\text{kJ g}^{-1}$ ?

- (A) 30.41
- (B) 36.10
- (C) 44.60
- (D) 47.79

**Question 5 (1 mark)**

Ethanoic acid ( $\text{CH}_3\text{COOH}$ ) is a weak acid with  $K_a = 1.74 \times 10^{-5}$  at  $25^\circ\text{C}$ . 1.00 L of a solution is prepared by dissolving 0.100 mol of ethanoic acid in distilled water.

What is the pH of the resulting solution?

- (A) 2.00
- (B) 2.10
- (C) 2.80
- (D) 2.88

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NESA STUDENT NUMBER

**Question 6 (1 mark)**

The table lists some properties of the straight-chained carbon compounds **W**, **X**, **Y** and **Z**.

| Compound | Reactivity in bromine water | Solubility in water |
|----------|-----------------------------|---------------------|
| <b>W</b> | Rapidly decolourises        | Insoluble           |
| <b>X</b> | Unreactive                  | Insoluble           |
| <b>Y</b> | Unreactive                  | Soluble             |
| <b>Z</b> | Unreactive                  | Partially soluble   |

Which row below best identifies the compounds **W**, **X**, **Y** and **Z**?

|     | <b>W</b>                      | <b>X</b>                      | <b>Y</b>                         | <b>Z</b>                         |
|-----|-------------------------------|-------------------------------|----------------------------------|----------------------------------|
| (A) | C <sub>3</sub> H <sub>6</sub> | C <sub>3</sub> H <sub>8</sub> | CH <sub>3</sub> OH               | C <sub>4</sub> H <sub>9</sub> OH |
| (B) | C <sub>3</sub> H <sub>8</sub> | C <sub>3</sub> H <sub>6</sub> | CH <sub>3</sub> OH               | C <sub>4</sub> H <sub>9</sub> OH |
| (C) | C <sub>3</sub> H <sub>6</sub> | C <sub>3</sub> H <sub>8</sub> | C <sub>4</sub> H <sub>9</sub> OH | CH <sub>3</sub> OH               |
| (D) | C <sub>3</sub> H <sub>8</sub> | C <sub>3</sub> H <sub>6</sub> | C <sub>4</sub> H <sub>9</sub> OH | CH <sub>3</sub> OH               |

**Question 7 (1 mark)**

The table below shows some data on four acid-base indicators.

| Indicator name  | pH range | Colour range |
|-----------------|----------|--------------|
| methyl orange   | 3.1–4.4  | red–yellow   |
| bromophenol red | 3.8–5.4  | yellow–blue  |
| methyl red      | 4.2–6.3  | red–yellow   |
| phenol red      | 6.8–8.4  | yellow–red   |

A solution, labelled **X**, is made by mixing equal volumes of 0.0002 mol L<sup>-1</sup> sodium hydroxide and 0.0004 mol L<sup>-1</sup> nitric acid.

Which indicator shown would turn pure yellow when added to a sample of **X**?

- (A) Methyl orange
- (B) Bromophenol red
- (C) Methyl red
- (D) Phenol red

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**Question 8 (1 mark)**

The following organic compounds all have five carbons:

**A:** 2,2-dimethylbutane

**B:** 2-methylpentane

**C:** 4-methylpent-2-ene

Which of the following is most likely to be correct?

- (A) Compound **A** has a lower melting point than compound **B**.
- (B) Compound **A** has a higher melting point than compound **B**.
- (C) Compound **B** and compound **C** are functional group isomers.
- (D) Compound **A** and compound **B** are positional isomers.

**Question 9 (1 mark)**

A 3.42 g sample of solid  $\text{Mg}(\text{OH})_2$  is added to a 200.0 mL solution of  $0.600 \text{ mol L}^{-1} \text{ HCl}$ .

What is the final pH of the final mixture?

- (A) 0.51
- (B) 1.2
- (C) 1.9
- (D) 2.6

**Question 10 (1 mark)**

A 50.0 mL sample of  $4.0 \times 10^{-3} \text{ M AgNO}_3$  solution is added to 50.0 mL of  $1.0 \times 10^{-3} \text{ M KIO}_3$  solution at  $25^\circ\text{C}$ .

The  $K_{sp}$  value for  $\text{AgIO}_3$  at  $25^\circ\text{C}$  is  $3.2 \times 10^{-8}$ .

Which row of the following table correctly predicts if  $\text{AgIO}_3$  will form a precipitate and the reason for this?

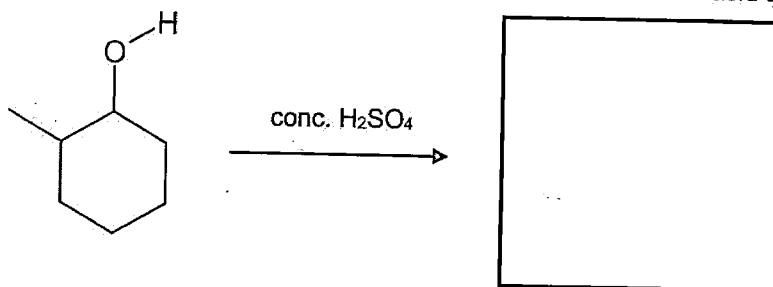
|     | Will a precipitate form? | Reason            |
|-----|--------------------------|-------------------|
| (A) | Yes                      | $Q_{sp} < K_{sp}$ |
| (B) | Yes                      | $Q_{sp} > K_{sp}$ |
| (C) | No                       | $Q_{sp} < K_{sp}$ |
| (D) | No                       | $Q_{sp} > K_{sp}$ |

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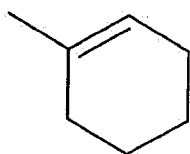
**Question 11 (1 mark)**

2-methylcyclohexanol is an alcohol and undergoes a dehydration reaction with sulfuric acid as shown.

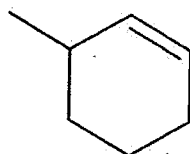


What is the major organic product of this reaction?

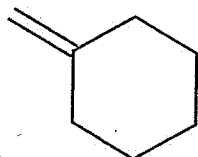
(A)



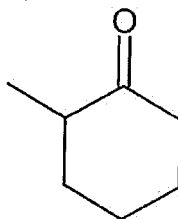
(B)



(C)

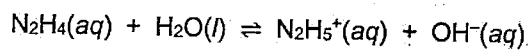


(D)



**Question 12 (1 mark)**

Consider the following equilibrium for a weak base N<sub>2</sub>H<sub>4</sub>.



A 0.75 mol sample of N<sub>2</sub>H<sub>4</sub> is placed in a 1.00 L flask. After the system comes to equilibrium, the flask contains 0.45 mol N<sub>2</sub>H<sub>5</sub><sup>+</sup>.

The value of K<sub>b</sub> for weak base N<sub>2</sub>H<sub>4</sub> is:

(A) 0.27

(B) 0.68

(C) 1.48

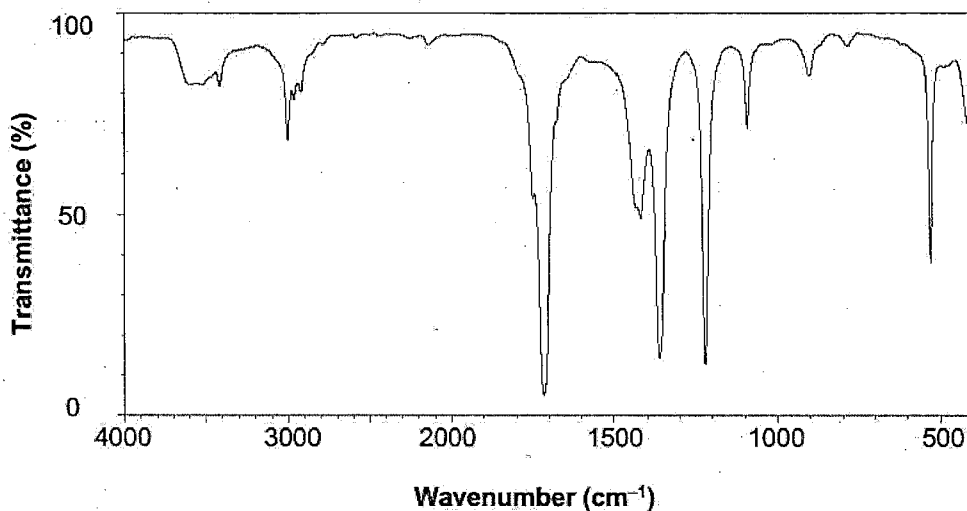
(D) 3.70

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**Question 13 (1 mark)**

The diagram shows the infrared spectrum of a compound.



Which compound was analysed?

- (A) Propane
- (B) Propan-1-ol
- (C) Propanal
- (D) Propan-2-one

**Question 14 (1 mark)**

A food critic noticed their specialty chicken dish was extremely bland and requested that analysis be carried out to measure the sodium chloride concentration. This was achieved by dissolving the food sample in water and the chloride ion being precipitated by adding an excess of silver nitrate solution. The precipitate was washed and dried.

If the mass of the food sample was 10.0 g and the final precipitate had a mass of 0.188 g, what is the percentage of sodium chloride in the food?

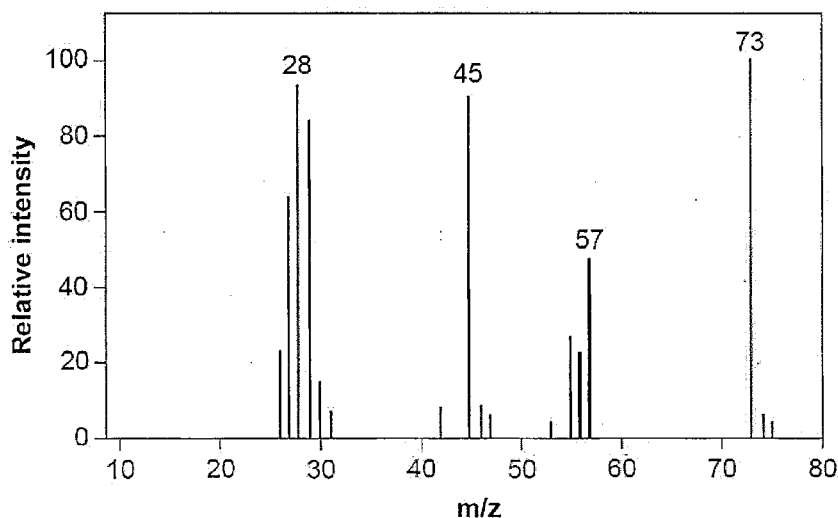
- (A) 1.88%
- (B) 0.766%
- (C) 0.465%
- (D) 0.220%

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**Question 15 (1 mark)**

A simplified mass spectrum of butanoic acid is shown.

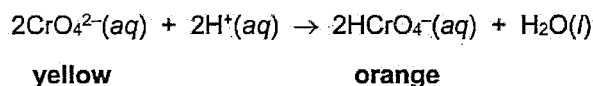


The peak with the mass-to-charge ratio,  $m/z$ , of 73:

- (A) has the greatest mass-to-charge ratio.
- (B) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}^-$ .
- (C) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}$ .
- (D) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}^+$ .

**Question 16 (1 mark)**

Silver chromate ( $\text{Ag}_2\text{CrO}_4$ ) is a yellow solid that dissolves in water to produce a yellow saturated solution containing  $2.34 \times 10^{-2} \text{ g L}^{-1}$  of silver chromate. The molar mass of silver chromate is  $331.8 \text{ g mol}^{-1}$ . The chromate ions react with hydrogen ions as shown below:



The molar concentration of chromate ions in a saturated solution of silver chromate in  $0.183 \text{ mol L}^{-1}$  silver nitrate is:

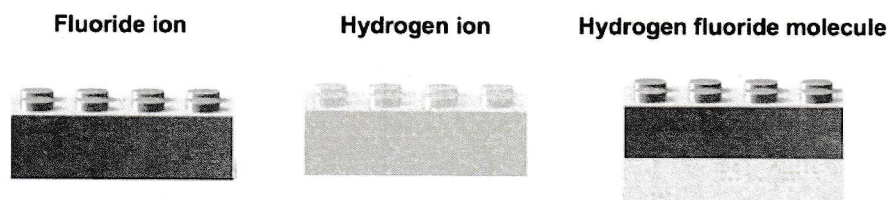
- (A)  $1.40 \times 10^{-12} \text{ g L}^{-1}$
- (B)  $4.19 \times 10^{-11} \text{ g L}^{-1}$
- (C)  $7.05 \times 10^{-3} \text{ g L}^{-1}$
- (D)  $2.34 \times 10^{-2} \text{ g L}^{-1}$

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Use the following information to answer questions 17–18.

A student was required to construct a model to help others visualise the composition of a solution of 0.1 M hydrofluoric acid. They were told this acid had a  $K_a$  value of  $6.7 \times 10^{-4}$ . The student had access to children's building blocks of various colours, like those shown in the image. The student counted out 25 dark blocks and 25 light blocks and created the following key.



The student used the  $K_a$  value provided to determine, via calculation, the appropriate number of dark and light blocks to join to make the number of 'HF molecules' in the model consistent with the  $K_a$  value for the acid.

After joining the calculated number of blocks, they finally placed all single and joined blocks into a large beaker. This was presented as the final model.

**Question 17 (1 mark)**

How many separate dark blocks should the student have placed into the beaker to make the model show the correct percent ionisation for the acid?

- (A) 2
- (B) 6
- (C) 12
- (D) 23

**Question 18 (1 mark)**

Other students assessed the model and were asked to state some of its limitations based on the aim of the modelling exercise. Four statements made are shown below.

- I. The model should have included some different coloured blocks to represent the proportion of hydroxide ions also present in the solution.
- II. The building blocks used were too big to represent actual ions and molecules.
- III. It did not include the most common chemical present in the 0.1 M HF being modelled.
- IV. The true nature of the hydrogen ion in aqueous solution is not shown by the model.

Given the aim of the exercise, which of the above statements are significant limitations of the final model presented?

- (A) I and II
- (B) I and IV
- (C) II and III
- (D) III and IV

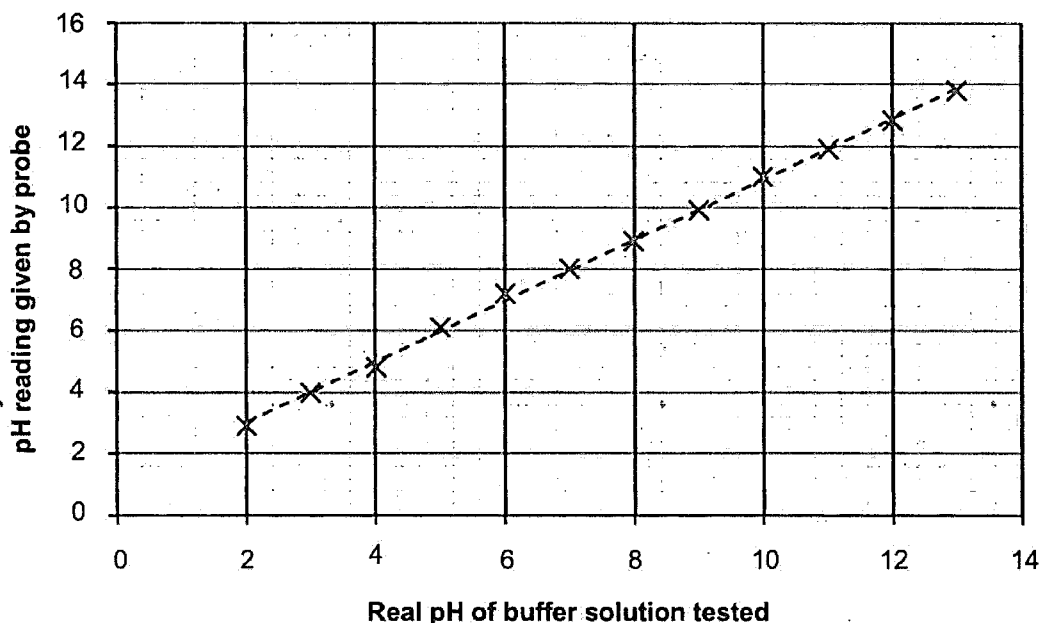
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**Question 19 (1 mark)**

A pH probe is used to measure the pH of several buffers of known pH values to determine if the meter required calibration. The graph below shows the pH readings given by the probe for each of the known (real) pH values of each buffer.

**Calibration graph for a pH probe**

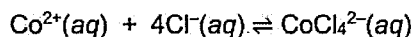


Which of the following statements is true, based on the data provided in the calibration graph shown?

- (A) Using the probe without calibration will affect the reliability of data it gathers.
- (B) If the probe were to be placed in a solution with  $[\text{OH}^-] = [\text{H}^+]$  it would read 6.
- (C) The equivalent point in a titration, as determined using this probe, will occur approximately one pH unit lower the actual equivalence point.
- (D) The  $[\text{H}^+]$  of a solution, as determined from its pH measured using this probe, will be 10 times lower than the actual  $[\text{H}^+]$  in the same solution.

**Question 20 (1 mark)**

A student performed an experiment to determine  $K_{\text{eq}}$  for the following reaction:



The student adds a 0.253 g sample of  $\text{CoCl}_2$  to a solution of 6.0 M  $\text{NaCl}$ . The value of  $K_{\text{eq}}$  is  $4.8 \times 10^{-4}$ .

Which of the following is the closest to the percentage of cobalt-containing species that is present as the blue  $\text{CoCl}_4^{2-}$  ion?

- (A) 38%
- (B) 46%
- (C) 54%
- (D) 62%

**END OF SECTION 1**

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NESA STUDENT NUMBER

## Section 2A

20 marks

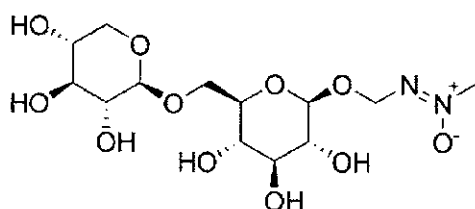
- Attempt Questions 1–4.
- If you use the extra space provided at the back of this exam, clearly indicate which question you are answering.

Marks

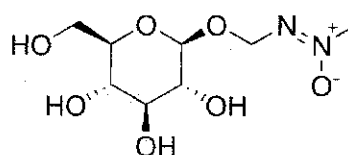
### Question 1 (3 marks)

Cycad fruits have been part of the staple diet of many Indigenous Australian peoples. However, these fruits contain toxins such as cycasin and macrozamin, which are known to cause long-term neurological damage.

3



macrozamin



cycasin

For safe consumption, the crushed powder from the cycad seeds was placed inside a woven bag. It was noted that running water was more efficient than still water in removing the toxins.

With reference to structural formulae and relevant chemical principles, justify the above observation used to remove the toxins.

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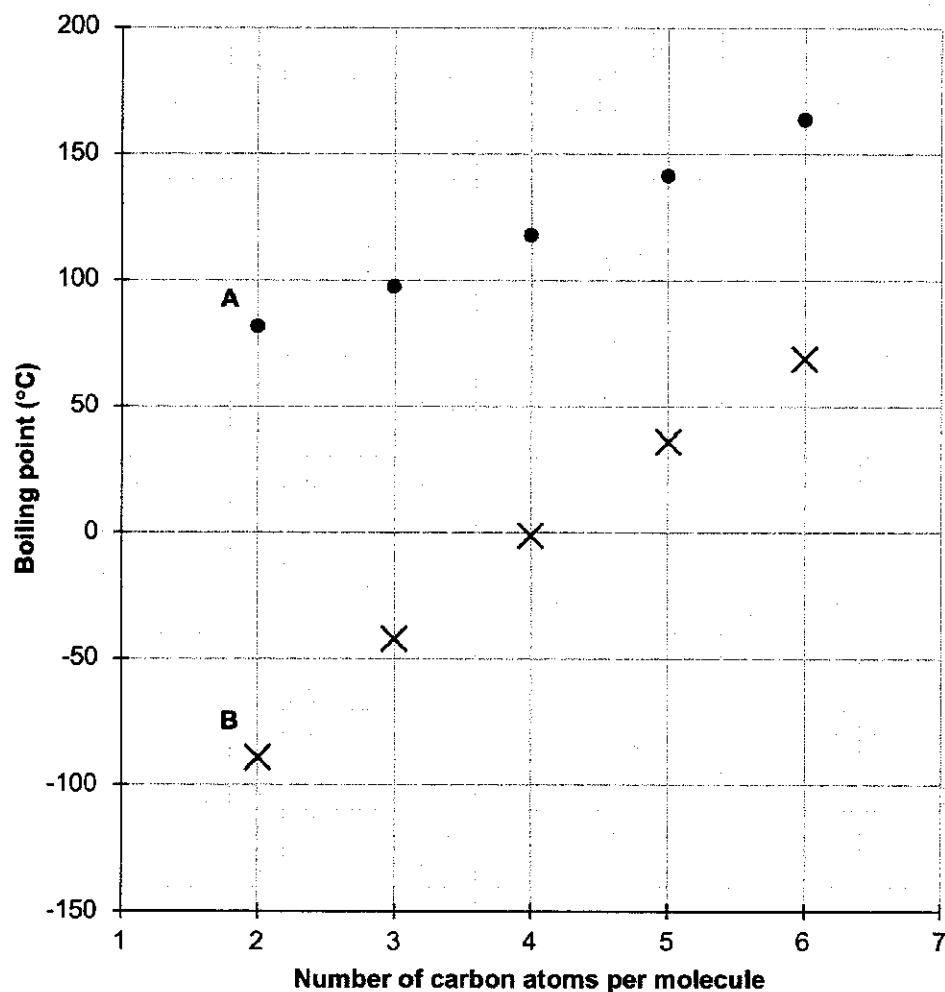
NESA STUDENT NUMBER

Marks

**Question 2 (5 marks)**

The graph below shows the boiling points for two different homologous series of organic compounds, straight chained alkanes and straight chained alkane nitriles ( $R-C\equiv N$ ).

The two homologous series are labelled on the graph as **A** and **B**.



Question 2 continued on the next page

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NESA STUDENT NUMBER

**Marks**

Question 2 continued...

Identify the label of **A** and **B** as either alkanes or alkane nitriles and explain the patterns of the boiling points shown in the graph.

**5**

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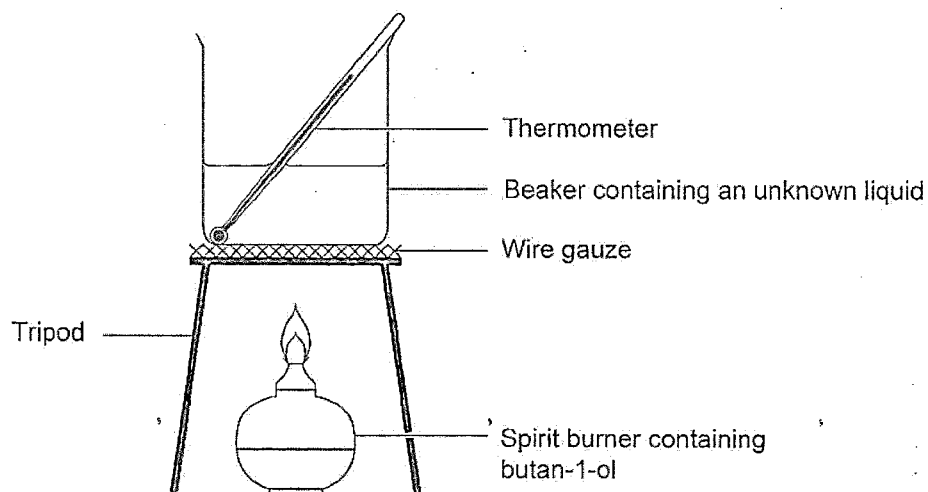
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NESA STUDENT NUMBER

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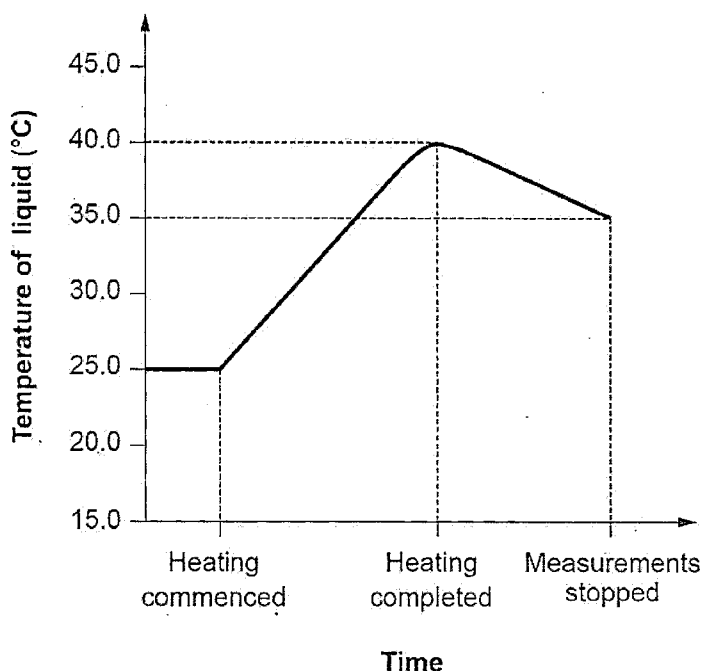
**Question 3 (5 marks)**

A 0.375 g sample of butan-1-ol is burnt to raise the temperature of 150 g of an unknown liquid, using the apparatus shown in **Figure 1**.



**Figure 1** – Apparatus used for the investigation

The temperature changes in the investigation are graphed in **Figure 2**.



**Figure 2** – Changes in the temperature of the unknown liquid during the investigation

**Question 3 continues on the next page**

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NESA STUDENT NUMBER

Marks

Question 3 continued...

- (a) The student noted that there was some soot on the base of the beaker.

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Outline a reason for this observation.

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- (b) Based on prior testing, it was determined that 35.0% of the heat produced by the combustion of butan-1-ol is lost to the surroundings.

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Using the information provided, calculate the specific heat capacity of the liquid, given that the heat of combustion of butan-1-ol is  $2676 \text{ kJ mol}^{-1}$ .

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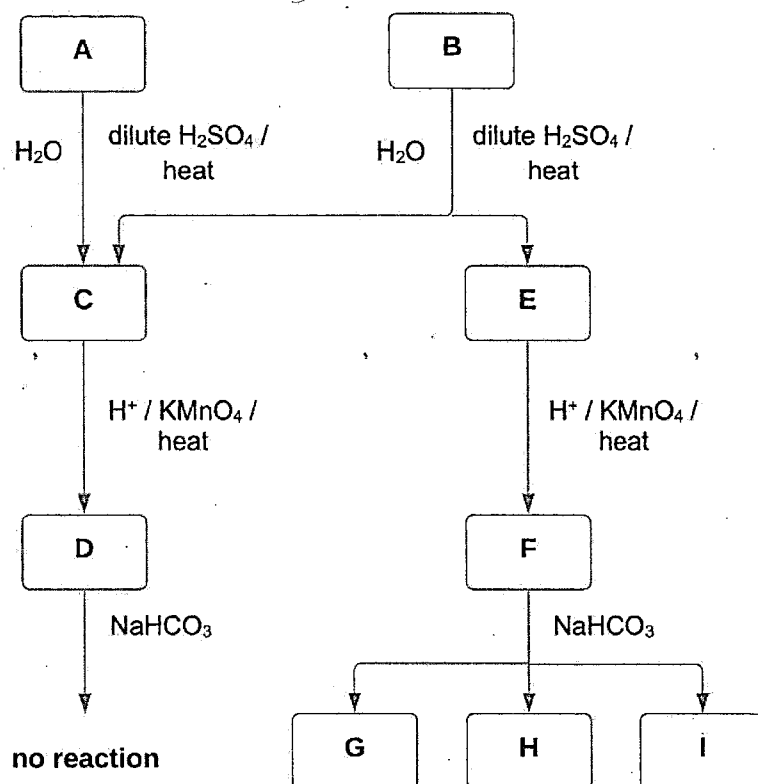
Marks

**Question 4 (7 marks)**

Two different compounds **A** and **B** are isomers with the molecular formula,  $C_4H_8$ .

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The flowchart below shows reactions involving six different compounds (**A** to **I**).



Question 4 continued on the next page

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NESA STUDENT NUMBER

**Marks**

Question 3 continued...

- (a) The student noted that there was some soot on the base of the beaker.

**1**

Outline a reason for this observation.

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- (b) Based on prior testing, it was determined that 35.0% of the heat produced by the combustion of butan-1-ol is lost to the surroundings.

**4**

Using the information provided, calculate the specific heat capacity of the liquid, given that the heat of combustion of butan-1-ol is  $2676 \text{ kJ mol}^{-1}$ .

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NESA STUDENT NUMBER

## Section 2B

40 marks

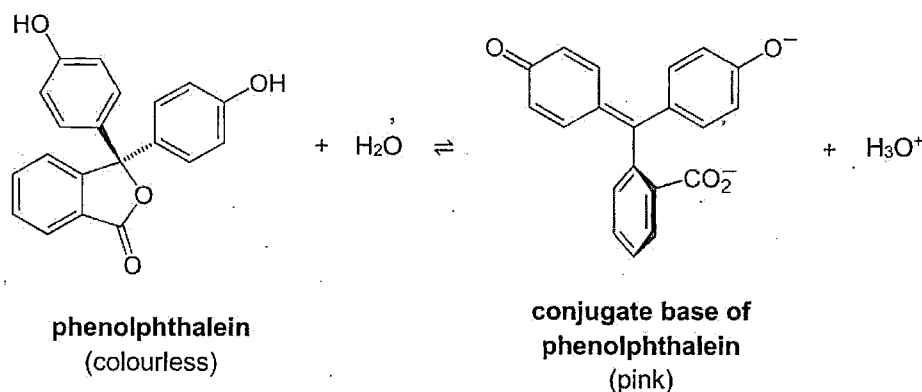
- Attempt Questions 1–8.
- If you use the extra space provided at the back of this exam, clearly indicate which question you are answering.

Marks

### Question 1 (3 marks)

Phenolphthalein is a common acid-base indicator with a  $K_a$  of  $3.0 \times 10^{-10}$  at  $25^\circ\text{C}$ .

When dissolved in water, it undergoes dissociation as shown in the equation below.



- (a) Distinct colour change for phenolphthalein occurs when half of the indicator is phenolphthalein, and the other half is its conjugate base. 1

Given that:

$$\text{pH} = \text{p}K_a + \log_{10} \frac{[\text{A}^-]}{[\text{HA}]}$$

calculate the pH at which phenolphthalein shows distinct colour change.

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Question 1 continues on the next page

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NESA STUDENT NUMBER

Marks

Question 1 continued...

- (b) The change of colour for phenolphthalein occurs over a limited range of pH and falls within  $\pm 1.00$  of the  $pK_a$  value.

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In an experiment, a solution of a weak acid was titrated against a solution of strong base. It was found that the equivalence point occurred at a pH of 9.30.

Explain if phenolphthalein is a suitable indicator for this titration.

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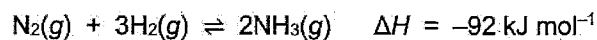
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Marks

**Question 2 (4 marks)**

The Haber Process is a method by which ammonia is produced commercially.

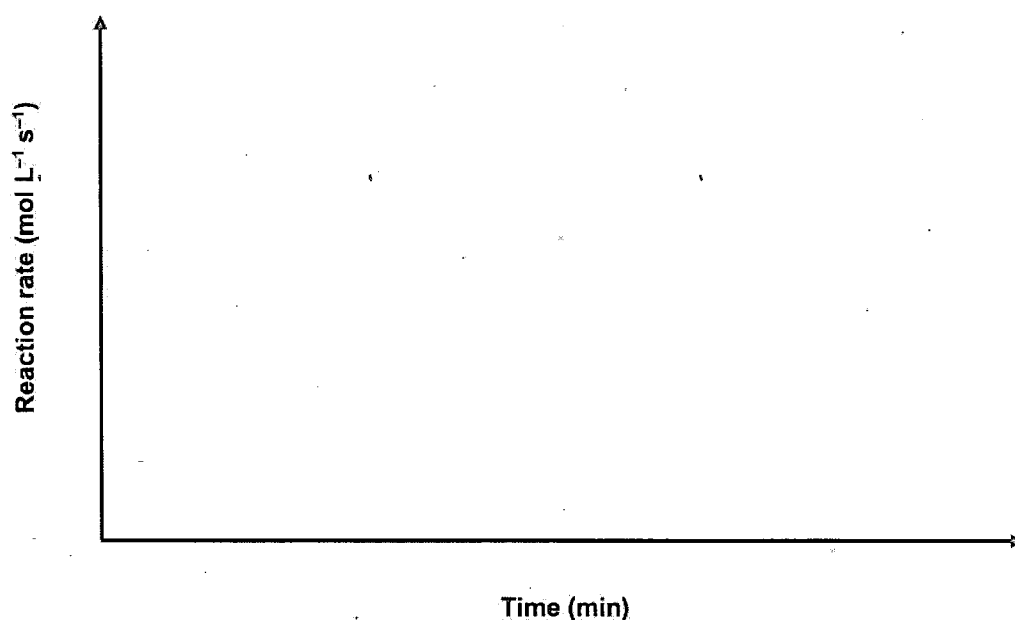
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A vessel contains all three gases at equilibrium.

The vessel is at equilibrium initially, then after 1 minute, the temperature of the vessel is increased. The system re-establishes equilibrium at the 2-minute mark.

Draw a labelled diagram showing the reaction rate-time graph for the above conditions described. Explain with reference to collision theory and Le Chatelier's Principle.



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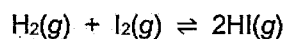
NESA STUDENT NUMBER

Marks

**Question 3 (5 marks)**

Hydrogen gas ( $H_2$ ) reacts with iodine gas ( $I_2$ ) in the following reversible reaction:

5



Joseph adds 0.200 mol of  $H_2$  and 0.200 mol of  $I_2$  to a 1.00 L closed container at  $400^\circ C$ . After 2 minutes, the system reaches equilibrium, and 0.316 mol of HI are found in the container.

At the 6-minute mark, how many moles of  $H_2$  gas must be added to the container to increase the amount of HI by 0.050 moles? The system re-establishes equilibrium after 7-minute mark.

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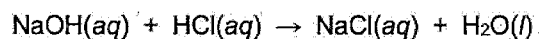
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**Question 4 (4 marks)**

The molar enthalpy of neutralisation of two reactions are given:

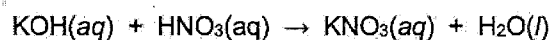
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**Reaction 1:**



$$\Delta H = -57.6 \text{ kJ mol}^{-1}$$

**Reaction 2:**



$$\Delta H = -57.6 \text{ kJ mol}^{-1}$$

Determine if the reaction between solutions of ammonia and hydrochloric acid will produce the same molar enthalpy of neutralisation value as **Reactions 1 and 2**.

Justify your response using your knowledge of acid-base chemistry and include an appropriate chemical equation.

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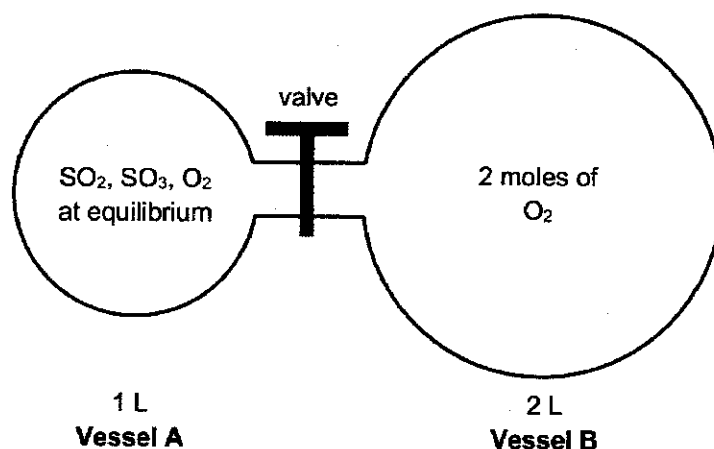
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NESA STUDENT NUMBER

Marks

**Question 5 (5 marks)**

The diagram below shows a closed system with vessels **A** and **B** connected with a small pipe with a valve at the centre. When the valve is closed gases of the two vessels cannot mix with each other. The volume of the small pipe with the valve is assumed to be negligible.



When the valve is closed, vessel **A** is at equilibrium denoted by  $K_{eq}$ . When the valve is opened, changes occur to the system in vessel **A**.

- (a) Show a relationship between the reaction quotient,  $Q$ , and  $K_{eq}$  when the valve is closed.

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- (b) Explain what happens to the equilibrium position when the valve is opened, using the most appropriate of the following principles.

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- Le Chatelier's principle
- Relative rates of the forward and back reactions
- Comparison of  $Q$  and  $K$

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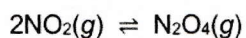
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**Question 6 (3 marks)**

Consider the following equilibrium system shown.



An experiment is conducted to understand the relationship between reversibility, temperature and entropy.



| Temperature (°C) | $\Delta G$ (kJ) | $K_{eq}$             |
|------------------|-----------------|----------------------|
| 100              | 8.4             | 0.067                |
| 0                | -9.2            | 58                   |
| -196             | -43.7           | $3.8 \times 10^{29}$ |

It is known that when  $\Delta G = 0$ , the reaction is at equilibrium ( $K_{eq} = 1$ ).

Explain the changes that occur when the sealed tube containing the brown  $\text{NO}_2$  gas is immersed in liquid nitrogen  $-196^\circ\text{C}$ , in terms of reversibility, enthalpy and entropy.

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NESA STUDENT NUMBER

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**Question 7 (5 marks)**

A student prepared two solutions by mixing aqueous sodium hydroxide and ethanoic acid.

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- **Solution A** was made by mixing together 25 mL 0.010 mol L<sup>-1</sup> aqueous sodium hydroxide with 50 mL 0.010 mol L<sup>-1</sup> ethanoic acid.
- **Solution B** was made by mixing together 25 mL 0.020 mol L<sup>-1</sup> aqueous sodium hydroxide with 50 mL 0.010 mol L<sup>-1</sup> ethanoic acid.

Compare the change in pH in **Solutions A** and **B** when a small amount of acid is added to the mixture using equilibrium principles and the Brønsted-Lowry theory.

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### Question 8 (11 marks)

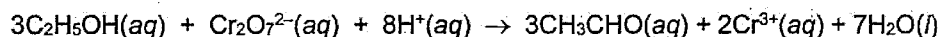
A manufacturer claims to have 30–32%(v/v) ethanol in most popular brand of red wine. The claim can be verified by determining the concentration of a sample of ethanol from the above red wine, using redox back titration method. The method is given below:

A sample of the red wine is diluted and then oxidised to ethanal using an excess of acidified potassium dichromate solution.

- Pipette 10.0 mL of wine into a 250.0 mL volumetric flask and add distilled water to make up to 250.0 mL and mix thoroughly.
- Pipette 20.0 mL of diluted wine into a conical flask.
- Add 20.0 mL aliquot of 0.040 mol L<sup>-1</sup> potassium dichromate solution, then add 10 mL of concentrated sulfuric acid (40%).
- Place stopper in flask loosely and heat in a 45°C water bath for 10 minutes then remove from heat.

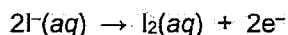
All of the ethanol (C<sub>2</sub>H<sub>5</sub>OH) in the wine sample is oxidised to ethanal (CH<sub>3</sub>CHO).

The flask is stoppered to prevent the ethanol being lost and the overall redox reaction is:

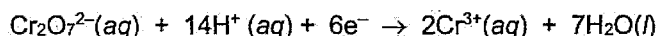


- Add 2 g of potassium iodide to the flask.

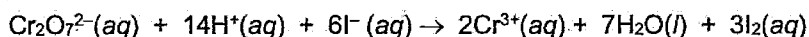
The excess dichromate ions will oxidise iodide ions:



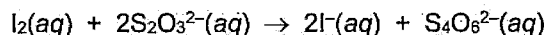
While the iodide ions reduce the dichromate ions to chromium (III) ions:



The overall redox reaction equation and the solution will look rather brown due to the presence of I<sub>2</sub>.



- Titrate the above solution containing iodine with 0.010 mol L<sup>-1</sup> sodium thiosulfate solution until the solution clears:



Thiosulfate ions act as a reducing agent, reducing I<sub>2</sub> to colourless I<sup>-</sup> and thereby removing the brown colour from the solution.

- Record the volume of sodium thiosulfate (the titre).
- Repeat procedure on several samples of the diluted wine until concordant titres are achieved.

Question 8 continued on the next page

NESA STUDENT NUMBER

Question 8 continued...

The results of the experiment are shown below.

| Sample | Volume of sodium thiosulfate (mL) |
|--------|-----------------------------------|
| A      | 31.92                             |
| B      | 29.63                             |
| C      | 29.61                             |
| D      | 29.62                             |

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Support your answer at least one relevant equation and all relevant calculations.

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**Question 8 continued on the next page**

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NESA STUDENT NUMBER

Marks

Question 8 continued...

- (b) A NBHS group of students performed some research after the previous volumetric analysis and has found the following pieces of information. They used the findings to alter the volumetric titration.

An extract from the students' logbook is shown below:

**Volumetric analysis of ethanol in a given sample of the red wine bottle**

**Procedure:**

- Volume of wine extracted – 50.00 mL
- Wine diluted to 500.0 mL
- 20.00 mL aliquot pipetted to a conical flask, titrated against 0.5022 M  $\text{KMnO}_4$  solution which is a very powerful oxidising agent.

**Observations (chemicals):**

- Wine diluted till clear fluid but  $\text{MnO}_4^-$  ions are intensely purple

**Observations (titrations):**

- **Titration #1** – Add first drop of titrant, contents of reaction flask turn pink immediately, then after a few seconds turn clear. Add a second drop of titrant, contents turn pink and immediately turns clear. Add titrant in a slow stream, contents remain a very slight pink colour, stop stream - contents turn clear immediately. Bubbles of carbon dioxide observed on the surface of the solution. Titrant added dropwise until colour change is persistent for 10 seconds.
- **Titration #2** – as for titration #1
- **Titration #3** – titrant added as a stream initially. Contents turn from pink immediately before turning clear. Brown solid observed on the bottom of the flask.

Careful research into the above observations led to the following findings.

- In the presence of a very powerful oxidising agent, ethanoic acid oxidises to carbon dioxide, thus inhibiting the reaction of  $\text{KMnO}_4$  with ethanol.
- $\text{Mn}^{2+}$  may be present as well and it acts as a catalyst, thus can cause an incomplete reduction of  $\text{MnO}_4^-$  ions and the reduction product is  $\text{MnO}_2$ , a brown suspension.

Question 8 continued on the next page

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NESA STUDENT NUMBER

Marks

Question 8 continued...

Using the observations on the previous page, suggest reasons as to why the data collected may or may not be accurate.

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In your response, include improvements to the experimental design.

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END OF SECTION 2B

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NESA STUDENT NUMBER

## Section 2C

20 marks

- Attempt Questions 1–4.
- If you use the extra space provided at the back of this section, clearly indicate which question you are answering.

Marks

### Question 1 (6 marks)

An unknown compound, **X**, is analysed and shown to have a composition of 69.7% carbon, 18.6% oxygen and the remainder is hydrogen. The following incomplete mass spectrometry, infrared spectroscopy data and chemical test data was collected.

#### Spectral data

| Mass spectrum                                                                                                                                 | Infrared spectrum                                                                                                                                                   |
|-----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• M+1 peak at <math>m/z = 87.0</math></li> <li>• Fragment ion peak at <math>m/z = 28</math></li> </ul> | <ul style="list-style-type: none"> <li>• Sharp peak at <math>1705\text{ cm}^{-1}</math></li> <li>• No broad peak around <math>3200\text{ cm}^{-1}</math></li> </ul> |

#### Chemical test data

- When Fehling's reagent (contains  $\text{Cu}^{2+}$  ions) is added to a sample of **X** in a test tube, no colour change is observed when the test tube is placed in a warm water bath.
- When Tollen's reagent (contains  $\text{Ag}^+$  ions) is added to a sample of **X** in a test tube, no silver mirror is produced.

Question 1 continues on the next page

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NESA STUDENT NUMBER

Marks

Question 1 continued...

- (a) Calculate the molecular formula of compound X.

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- (b) Using the data above, identify two possible isomers for compound X. Justify your response by including an explanation proving your two possible isomers and disproving one other isomer.

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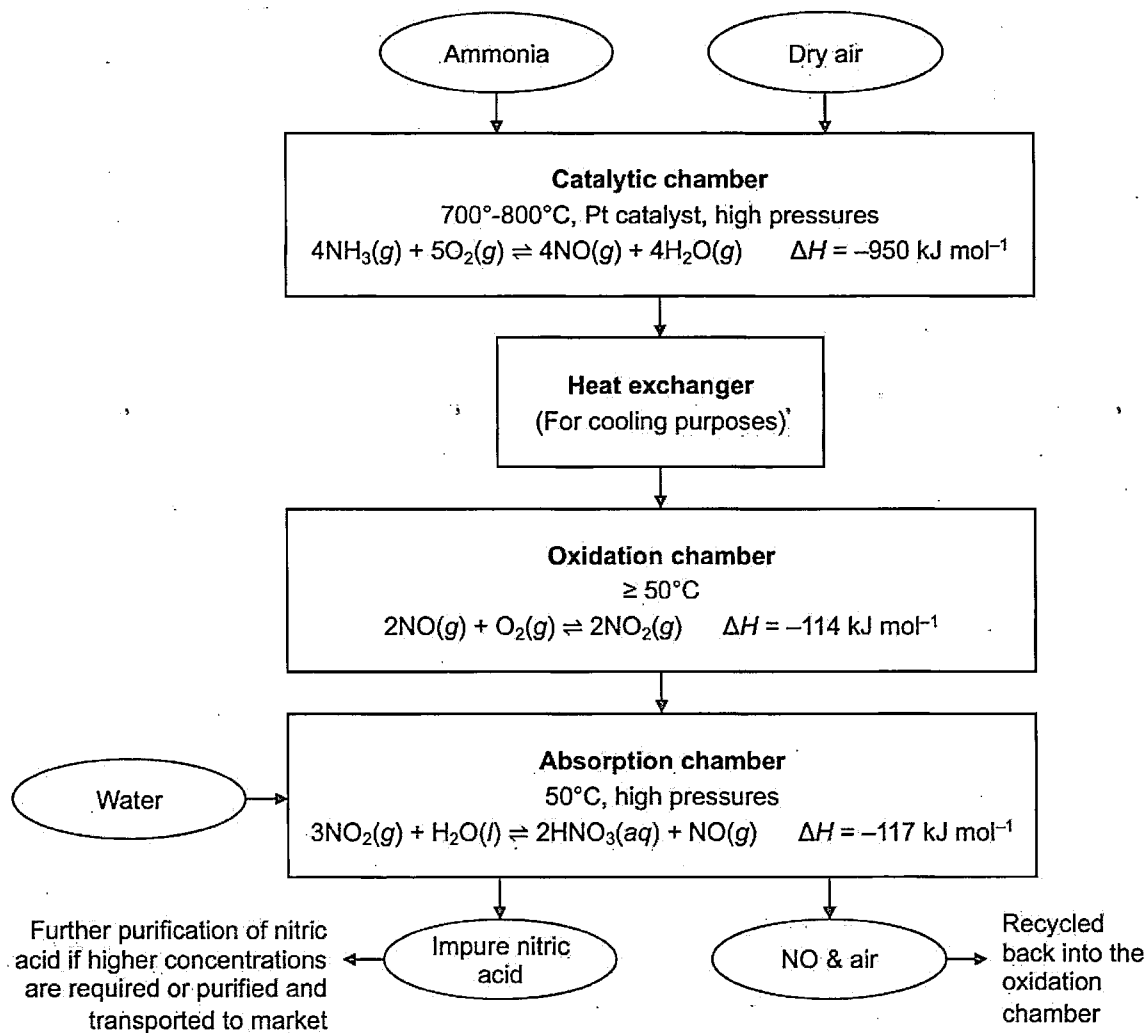
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Marks

**Question 2 (4 marks)**

The flow chart summarises an industrial process for the synthesis of nitric acid.



Question 2 continues on the next page

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**Marks**

Question 2 continued...

Explain THREE factors that may have been considered in the design of this industrial process. Make specific reference to the flow chart.

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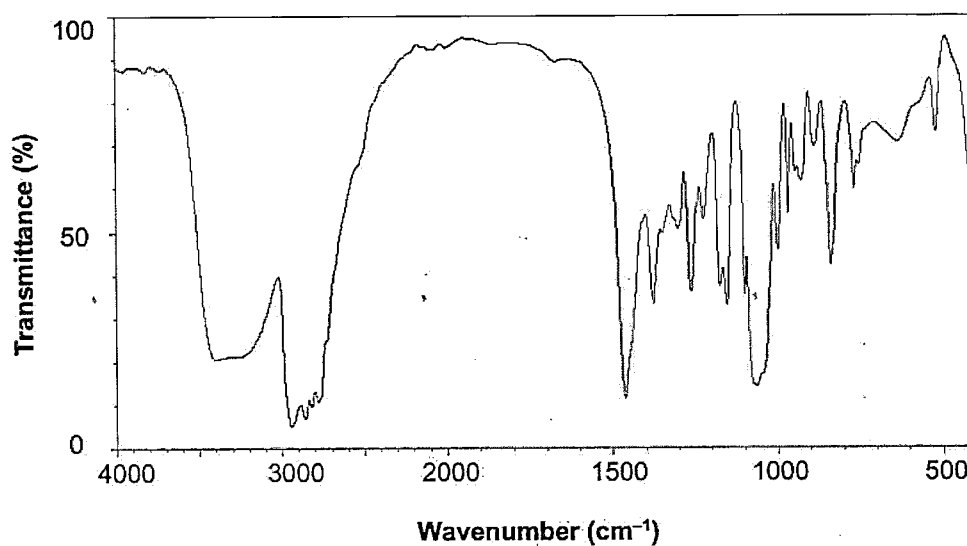
**Question 3 (7 marks)**

A chemist discovered a bottle simply labelled ' $C_5H_{13}NO$ '.

7

To confirm the molecular structure of the contents of the bottle, a sample was submitted for analysis by infrared spectroscopy and  $^1H$  and  $^{13}C$  NMR spectroscopy. The resulting spectra are shown.

**Infrared spectrum**



Question 3 continues on the next page

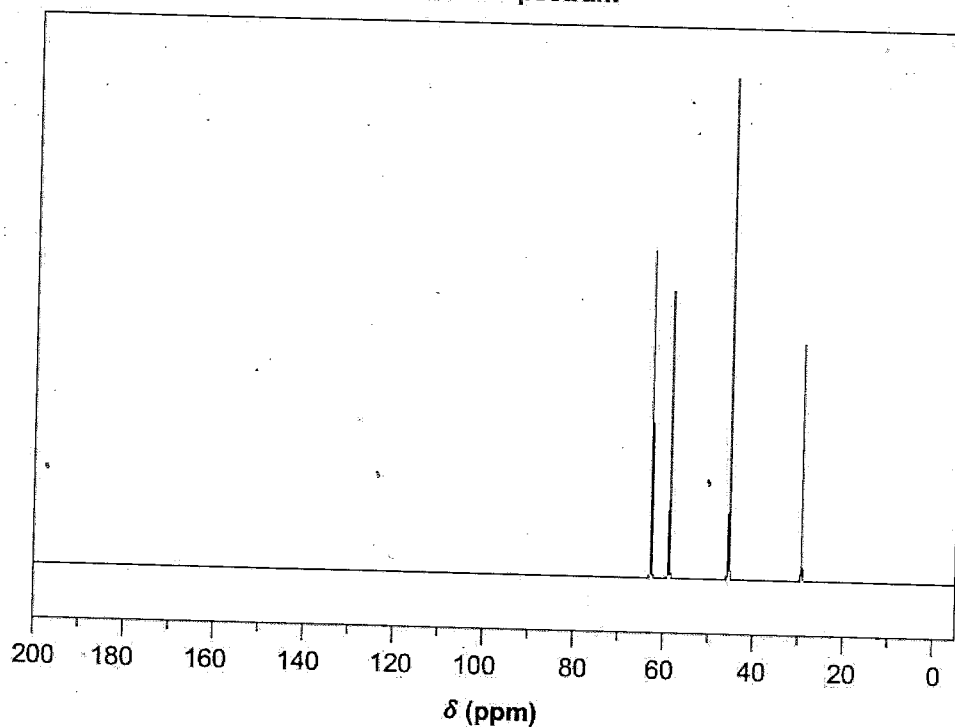
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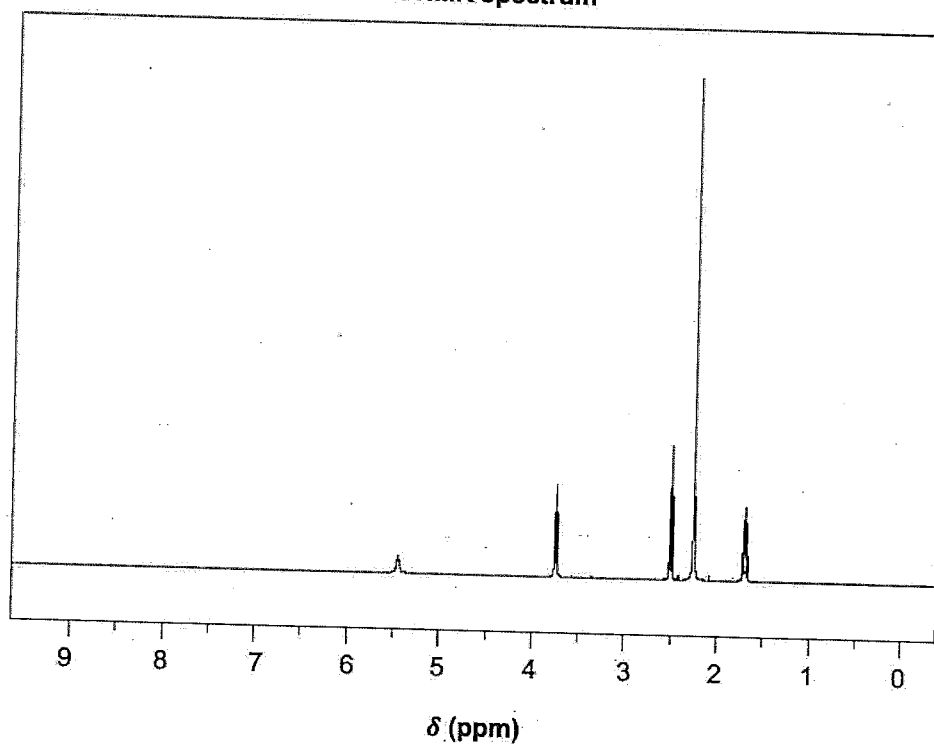
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Question 3 continued...

$^{13}\text{C}$  NMR spectrum



$^1\text{H}$  NMR spectrum



Question 3 continues on the next page

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NESA STUDENT NUMBER

Marks

Question 3 continued...

Data from the  $^1\text{H}$  NMR spectrum

| Chemical shift | Relative peak area | Splitting pattern |
|----------------|--------------------|-------------------|
| 5.45           | 1                  | singlet (1)       |
| 3.75           | 2                  | triplet (3)       |
| 2.50           | 2                  | triplet (3)       |
| 2.25           | 6                  | singlet (1)       |
| 1.70           | 2                  | quintet (5)       |

$^1\text{H}$  NMR chemical shift data

| Type of proton                                                 | $\delta$ (ppm) |
|----------------------------------------------------------------|----------------|
| $\text{Si}(\text{CH}_3)_4$ (TMS)                               | 0              |
| $\text{R}-\text{CH}_3$                                         | 0.7–1.3        |
| $\text{R}-\text{CH}_2-\text{R}$                                | 1.2–1.5        |
| $\text{R}-\text{CHR}_2$                                        | 1.5–2.0        |
| $\text{H}_3\text{C}-\text{CO}-$ (aldehydes, ketones or esters) | 2.0–2.5        |
| $-\text{CH}-\text{CO}-$ (aldehydes, ketones or esters)         | 2.1–2.6        |
| $\text{H}_3\text{C}-\text{O}-$ (alcohols or esters)            | 3.2–4.0        |
| $-\text{CH}-\text{O}-$ (alcohols or esters)                    | 3.3–5.1        |
| $\text{R}_2-\text{CH}_2-\text{O}-$ (alcohols or esters)        | 3.5–5.0        |
| $\text{R}-\text{OH}$                                           | 1–6            |
| $\text{R}_2\text{C}=\text{CHR}$ (alkene)                       | 4.5–7.0        |
| $\text{R}-\text{CHO}$ (aldehyde)                               | 9.4–10.0       |
| $\text{R}-\text{COOH}$                                         | 9.0–13.0       |

Question 3 continues on the next page

NESA STUDENT NUMBER:

Question 3 continued...

Justify your answer with reference to the information provided.

**Note:** The IUPAC name of the unknown compound is not required.

[illegible]

**Question 3 continues on the next page**

|  |  |  |  |  |  |  |  |  |  |
|--|--|--|--|--|--|--|--|--|--|
|  |  |  |  |  |  |  |  |  |  |
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NESA STUDENT NUMBER

Marks

Question 3 continued...

Handwriting practice lines consisting of 10 horizontal dotted lines.

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NESA STUDENT NUMBER

Marks

**Question 4 (3 marks)**

Calcium compounds such as calcium fluoride and calcium hydroxide are sparingly soluble in water.

**3**

To investigate the solubility of calcium fluoride in calcium hydroxide, a student prepared a saturated solution consisting of calcium fluoride, calcium hydroxide and water. It had a pH of 12.0.

The student noted that the  $K_{sp}$  of calcium fluoride at 25°C is  $3.45 \times 10^{-11}$ .

Calculate the molar solubility of calcium fluoride in the above solution at 25°C.

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**END OF SECTION 2C**

# Normanhurst Boys High School

## 2023 Year 12 Chemistry Trial Examination

### Sample Answers & Notes from the Marker

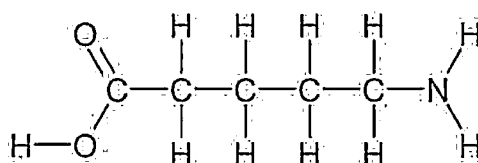
S1

#### Section 1 (20 marks)

#### Written Response

##### Question 1 (1 mark)

An organic compound shown below.



Which are the TWO functional groups present in this molecule?

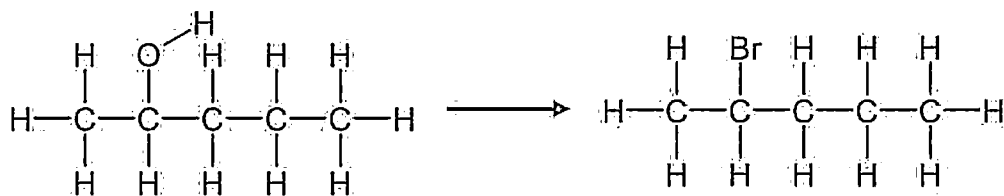
- (A) Amide and ester.
- (B) Amide and carbonyl.
- (C) Carboxyl and amine.
- (D) Alkene and amine.

##### Notes from the Marker

The two functional groups are on the ends of this molecule, the carboxyl group ( $-\text{COOH}$ ) and the amine group ( $-\text{NH}_2$ ).

##### Question 2 (1 mark)

Which of the following conditions would be appropriate to carry out the following synthesis?



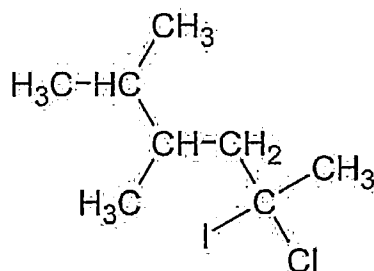
- (A)  $\text{Br}_2$  and UV light.
- (B)  $\text{Br}_2$  dissolved in water.
- (C)  $\text{HBr}$  and heat.
- (D)  $\text{HBr}$  and  $\text{ZnCl}_2$  catalyst.

##### Notes from the Marker

Substitution reaction of secondary alcohols with  $\text{HBr}$  and heat.

**Question 3 (1 mark)**

The structure of a compound is shown below.



What is the IUPAC name of this compound?

(A) 2-chloro-2-iodo-4,5-dimethylhexane

(B) 2-iodo-2-chloro-4,5-dimethylhexane

(C) 2,3-dimethyl-5-chloro-5-iodohexane

(D) 5-chloro-5-iodo-2,3-dimethylhexane

**Notes from the Marker**

Start with finding the longest carbon chain, in this case is hexane. Number the carbon chain so that each substituent is given a locant.

There are two possible ways to number the carbon chain:

- From left to right: 2, 3, 5, 5
- From right to left: 2, 2, 4, 5

First point of difference rule will then apply, where the assigned locants will be chosen based on the first substituent locant that has the lowest locant, hence: 2, 2, 4, 5.

The substituents should also be named in alphabetical order.

**Question 4 (1 mark)**

The heat of combustion of butan-1-ol is  $2676 \text{ kJ mol}^{-1}$ .

What is the value of the heat of combustion in  $\text{kJ g}^{-1}$ ?

(A) 30.41

(B) 36.10

(C) 44.60

(D) 47.79

**Notes from the Marker**

To convert  $\Delta H = 2676 \text{ kJ mol}^{-1}$  into  $\text{kJ g}^{-1}$ , divide  $\Delta H$  by the molar mass of butan-1-ol ( $\text{C}_4\text{H}_9\text{OH}$ ), since molar mass has units of  $\text{g mol}^{-1}$ .

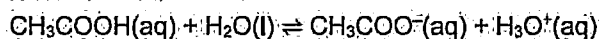
$$\begin{aligned}\frac{\Delta H}{MM} &= \frac{2676}{4(12.01) + 10(1.008) + 16.00} \\ &= 36.10 \text{ kJ g}^{-1}\end{aligned}$$

**Question 5 (1 mark)**

Ethanoic acid ( $\text{CH}_3\text{COOH}$ ) is a weak acid with  $K_a = 1.74 \times 10^{-5}$  at  $25^\circ\text{C}$ . 1.00 L of a solution is prepared by dissolving 0.100 mol of ethanoic acid in distilled water.

What is the pH of the resulting solution?

- (A) 2.00  
(B) 2.10  
(C) 2.80  
(D) 2.88

**Notes from the Marker:**

$$[\text{CH}_3\text{COOH}] = 0.100 \text{ M}$$

$$K_a = \frac{[\text{CH}_3\text{COO}^-][\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}]}$$

$$= \frac{[\text{H}_3\text{O}^+]^2}{[\text{CH}_3\text{COOH}]} \quad (\text{since } [\text{CH}_3\text{COO}^-] = [\text{H}_3\text{O}^+])$$

$$1.74 \times 10^{-5} = \frac{[\text{H}_3\text{O}^+]^2}{0.100}$$

$$[\text{H}_3\text{O}^+] = 1.319 \times 10^{-3} \text{ M}$$

$$\text{pH} = -\log_{10}(1.319 \times 10^{-3})$$

$$= 2.88$$

**Question 6 (1 mark)**

The table lists some properties of the straight-chained carbon compounds W, X, Y and Z.

| Compound | Reactivity in bromine water | Solubility in water |
|----------|-----------------------------|---------------------|
| W        | Rapidly decolourises        | Insoluble           |
| X        | Unreactive                  | Insoluble           |
| Y        | Unreactive                  | Soluble             |
| Z        | Unreactive                  | Partially soluble   |

Which row below best identifies the compounds W, X, Y and Z?

|     | W                      | X                      | Y                               | Z                               |
|-----|------------------------|------------------------|---------------------------------|---------------------------------|
| (A) | $\text{C}_3\text{H}_6$ | $\text{C}_3\text{H}_8$ | $\text{CH}_3\text{OH}$          | $\text{C}_4\text{H}_9\text{OH}$ |
| (B) | $\text{C}_3\text{H}_8$ | $\text{C}_3\text{H}_6$ | $\text{CH}_3\text{OH}$          | $\text{C}_4\text{H}_9\text{OH}$ |
| (C) | $\text{C}_3\text{H}_6$ | $\text{C}_3\text{H}_8$ | $\text{C}_4\text{H}_9\text{OH}$ | $\text{CH}_3\text{OH}$          |
| (D) | $\text{C}_3\text{H}_8$ | $\text{C}_3\text{H}_6$ | $\text{C}_4\text{H}_9\text{OH}$ | $\text{CH}_3\text{OH}$          |

### Notes from the Marker

- W is an alkene,  $C_3H_6$  (propene), which decolourises bromine water due to the presence of a double bond. It is also non-polar and insoluble in water.
- X is an alkane,  $C_3H_8$  (propane), so is unreactive in bromine water, as it is saturated. It is also non-polar and insoluble in water.
- Y is a small alcohol,  $CH_3OH$  (methanol), so is unreactive in bromine water, as it is saturated. It also contains a polar hydroxyl group with a very short hydrocarbon chain, so is soluble in water, since the hydroxyl group can form hydrogen bonds with water.
- Z is a longer chained alcohol,  $C_4H_9OH$ , so is unreactive in bromine water, as it is saturated. It also contains a polar hydroxyl group, which can form hydrogen bonds with water. However, the hydrocarbon chain is longer and is non-polar, hence decreasing the solubility of the alcohol, making it partially soluble.

### Question 7 (1 mark)

The table below shows some data on four acid-base indicators.

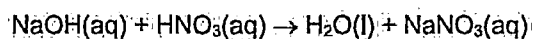
| Indicator name  | pH range | Colour range |
|-----------------|----------|--------------|
| methyl orange   | 3.1–4.4  | red–yellow   |
| bromophenol red | 3.8–5.4  | yellow–blue  |
| methyl red      | 4.2–6.3  | red–yellow   |
| phenol red      | 6.8–8.4  | yellow–red   |

A solution, labelled X, is made by mixing equal volumes of  $0.0002 \text{ mol L}^{-1}$  sodium hydroxide and  $0.0004 \text{ mol L}^{-1}$  nitric acid.

Which indicator shown would turn pure yellow when added to a sample of X?

- (A) Methyl orange  
(B) Bromophenol red  
(C) Methyl red  
(D) Phenol red

### Notes from the Marker



If 1 L of each reactant were added together:

$$\begin{aligned} n(NaOH) &= 0.0002 \times 1 \\ &= 0.0002 \text{ mol} \end{aligned}$$

$$\begin{aligned} n(HNO_3) &= 0.0004 \times 1 \\ &= 0.0004 \text{ mol} \end{aligned}$$

Due to 1:1 mole ratio, the left-over moles after the reaction:

$$\begin{aligned} n(H_3O^+) - n(OH^-) &= 0.0004 - 0.0002 \\ &= 0.0002 \text{ mol of } H_3O^+ \end{aligned}$$

$$\begin{aligned} [H_3O^+] &= n/V \\ &= 0.0002 / 2 \\ &= 0.0001 \text{ M} \\ pH &= -\log_{10}(0.0001) \\ &= 4 \end{aligned}$$

Hence, for the indicator to be pure yellow, it must be phenol red, as it will be yellow for  $pH < 6.8$ .

**Question 8 (1 mark)**

The following organic compounds all have five carbons.

A: 2,2-dimethylbutane

B: 2-methylpentane

C: 4-methylpent-2-ene

Which of the following is most likely to be correct?

- (A) Compound A has a lower melting point than compound B.
- (B) Compound A has a higher melting point than compound B.
- (C) Compound B and compound C are functional group isomers.
- (D) Compound A and compound B are positional isomers.

**Notes from the Marker**

The surface area of compound A is lower than the surface area of compound B, due to more branches. Compound A is therefore easier to pack together and has a higher melting point compared to compound B.

**Question 9 (1 mark)**

A 3.42 g sample of solid  $\text{Mg}(\text{OH})_2$  is added to a 200.0 mL solution of  $0.600 \text{ mol L}^{-1} \text{ HCl}$ .

What is the final pH of the final mixture?

- (A) 0.51
- (B) 1.2
- (C) 1.9
- (D) 2.6

**Notes from the Marker**

$$\begin{aligned} n(\text{Mg}(\text{OH})_2) &= \frac{m}{MM} \\ &= \frac{3.42}{24.31 + 2(16.00) + 2(1.008)} \\ &= 0.0586359428 \text{ mol} \\ \text{Mg}(\text{OH})_2(\text{aq}) &\rightarrow \text{Mg}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \\ n(\text{OH}^{-}) &= 0.0586359428 \times 2 \text{ (due to 1:2 mole ratio)} \\ &= 0.1172718856 \text{ mol} \end{aligned}$$

$$\begin{aligned} n(\text{HCl}) &= c \times V \\ &= 0.600 \times 0.200 \\ &= 0.1200 \text{ mol} \\ \text{Hence, H}^+ \text{ will be in excess:} \\ n(\text{H}^+)_{\text{excess}} &= 0.1200 - 0.1172718856 \\ &= 0.0027281144 \text{ mol} \\ [\text{H}^+] &= n/V \\ &= \frac{0.0027281144}{0.200} \\ &= 0.01364 \text{ M} \\ \text{pH} &= -\log_{10}(0.01364) \\ &= 1.9 \end{aligned}$$

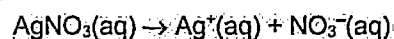
**Question 10 (1 mark)**

A 50.0 mL sample of  $4.0 \times 10^{-3}$  M  $\text{AgNO}_3$  solution is added to 50.0 mL of  $1.0 \times 10^{-3}$  M  $\text{KIO}_3$  solution at  $25^\circ\text{C}$ .

The  $K_{sp}$  value for  $\text{AgIO}_3$  at  $25^\circ\text{C}$  is  $3.2 \times 10^{-8}$ .

Which row of the following table correctly predicts if  $\text{AgIO}_3$  will form a precipitate and the reason for this?

|     | Will a precipitate form? | Reason            |
|-----|--------------------------|-------------------|
| (A) | Yes                      | $Q_{sp} < K_{sp}$ |
| (B) | Yes                      | $Q_{sp} > K_{sp}$ |
| (C) | No                       | $Q_{sp} < K_{sp}$ |
| (D) | No                       | $Q_{sp} > K_{sp}$ |

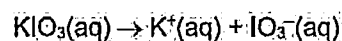
**Notes from the Marker**

$$n(\text{AgNO}_3) = 4.0 \times 10^{-3} \times 0.050$$

$$= 0.0002 \text{ mol}$$

$$[\text{Ag}^+]_{\text{mixed}} = \frac{0.0002}{0.100}$$

$$= 0.002 \text{ M}$$

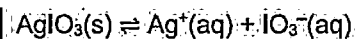


$$n(\text{KIO}_3) = 1.0 \times 10^{-3} \times 0.050$$

$$= 0.00005 \text{ mol}$$

$$[\text{IO}_3^-]_{\text{mixed}} = \frac{0.00005}{0.100}$$

$$= 0.0005 \text{ M}$$



$$Q_{sp} = [\text{Ag}^+][\text{IO}_3^-]$$

$$= (0.002)(0.0005)$$

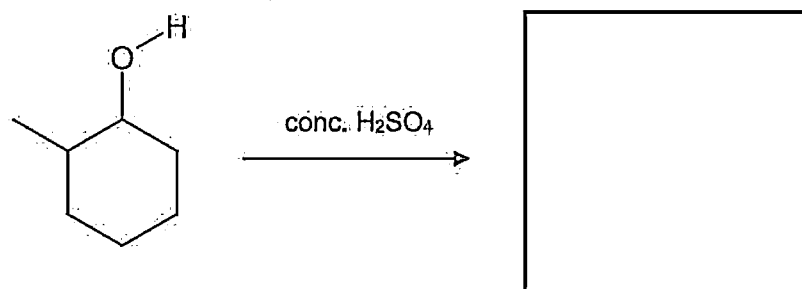
$$= 1.0 \times 10^{-6} \text{ M}$$

$$Q_{sp} > K_{sp}$$

Therefore, a precipitate will form.

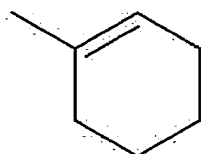
**Question 11** (1 mark)

2-methylcyclohexanol is an alcohol and undergoes a dehydration reaction with sulfuric acid as shown.

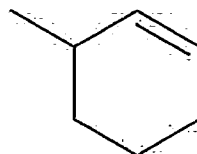


What is the major organic product of this reaction?

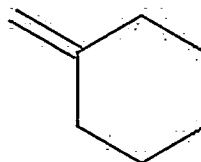
(A)



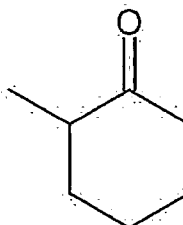
(B)



(C)



(D)



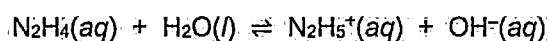
**Notes from the Marker**

In a dehydration reaction, the hydroxyl group and a hydrogen atom from an adjacent carbon is removed to form an alkene and water. This results in the products of option (A) and (B) that can be produced.

The major product, however, is the alkene with most carbons directly attached (i.e. most highly substituted alkene); hence option (A) is the major product.

**Question 12** (1 mark)

Consider the following equilibrium for a weak base  $\text{N}_2\text{H}_4$ .



A 0.75 mol sample of  $\text{N}_2\text{H}_4$  is placed in a 1.00 L flask. After the system comes to equilibrium, the flask contains 0.45 mol  $\text{N}_2\text{H}_5^+$ .

The value of  $K_b$  for weak base  $\text{N}_2\text{H}_4$  is:

(A) 0.27

(B) 0.68

(C) 1.48

(D) 3.70

**Notes from the Marker**

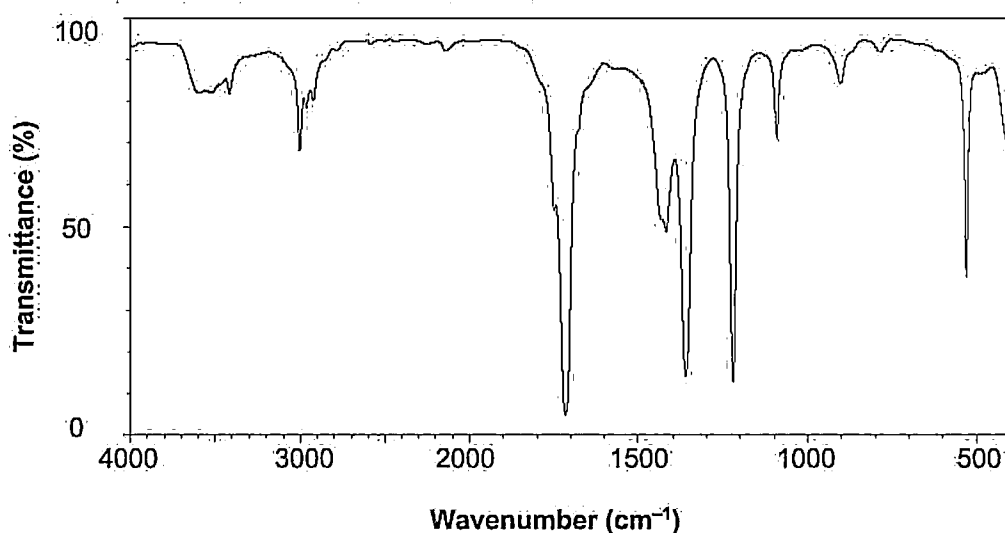
$$[\text{N}_2\text{H}_4] = \frac{0.75}{1.00}$$
$$= 0.75 \text{ M}$$

|             | $[\text{N}_2\text{H}_4]$ | $[\text{N}_2\text{H}_5^+]$ | $[\text{OH}^-]$ |
|-------------|--------------------------|----------------------------|-----------------|
| Initial     | 0.75                     | 0.00                       | 0.00            |
| Change      | -0.45                    | +0.45                      | +0.45           |
| Equilibrium | 0.30                     | 0.45                       | 0.45            |

$$K_b = \frac{[\text{N}_2\text{H}_5^+][\text{OH}^-]}{[\text{N}_2\text{H}_4]}$$
$$= \frac{(0.45)(0.45)}{0.30}$$
$$= 0.68$$

**Question 13 (1 mark)**

The diagram shows the infrared spectrum of a compound.



Which compound was analysed?

- (A) Propane
- (B) Propan-1-ol
- (C) Propanal
- (D) Propan-2-one

**Notes from the Marker**

The C-H bond is detected at  $\sim 3000 \text{ cm}^{-1}$  and the C=O bond is detected at  $\sim 1700 \text{ cm}^{-1}$ . These bonds are present in propanal and propan-2-one.

**Question 14** (1 mark)

A food critic noticed their specialty chicken dish was extremely bland and requested that analysis be carried out to measure the sodium chloride concentration. This was achieved by dissolving the food sample in water and the chloride ion being precipitated by adding an excess of silver nitrate solution. The precipitate was washed and dried.

If the mass of the food sample was 10.0 g and the final precipitate had a mass of 0.188 g, what is the percentage of sodium chloride in the food?

(A) 1.88%

(B) 0.766%

(C) 0.465%

(D) 0.220%

**Notes from the Marker:**

$$m(\text{AgCl}) = 0.188 \text{ g}$$

$$n(\text{AgCl}) = m/\text{MM}$$

$$= \frac{0.188}{107.9 + 35.45}$$

$$= 0.00131147541 \text{ mol}$$

$$n(\text{Cl}^-) = 0.00131147541 \text{ mol}$$

In NaCl:

$$n(\text{NaCl}) = 0.00131147541 \text{ mol}$$

$$m(\text{NaCl}) = n \times \text{MM}$$

$$= 0.00131147541 \times (22.99 + 35.45)$$

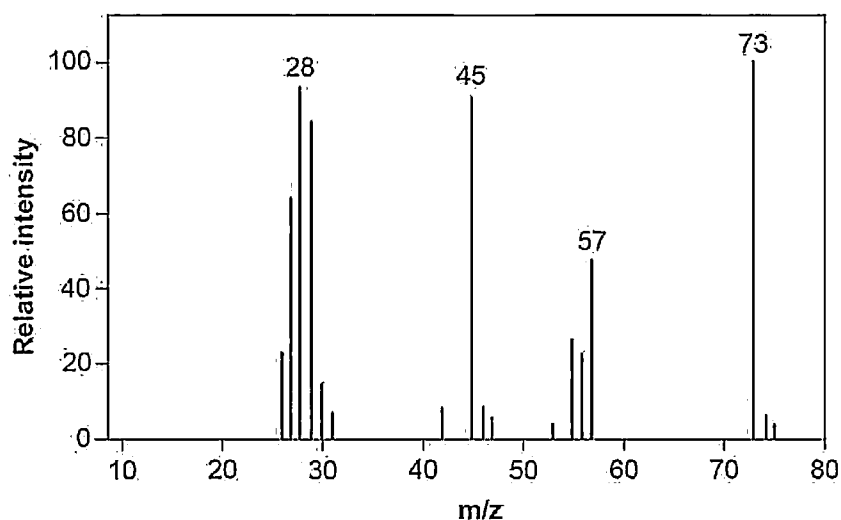
$$= 0.07664 \text{ g}$$

$$\%(\text{NaCl}) = \frac{0.07664}{10.0} \times 100$$

$$= 0.766\%$$

**Question 15** (1 mark)

A simplified mass spectrum of butanoic acid is shown.



The peak with the mass-to-charge ratio,  $m/z$ , of 73:

(A) has the greatest mass-to-charge ratio.

(B) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}^-$ .

(C) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}$ .

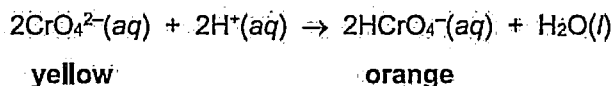
(D) represents the species  $\text{CH}_2\text{CH}_2\text{COOH}^+$ .

### Notes from the Marker

The  $\text{CH}_2\text{CH}_2\text{COOH}^+$  species has a mass of 73 and only positively charged species can be detected in a mass spectrometer.

### Question 16 (1 mark)

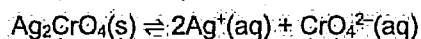
Silver chromate ( $\text{Ag}_2\text{CrO}_4$ ) is a yellow solid that dissolves in water to produce a yellow saturated solution containing  $2.34 \times 10^{-2} \text{ g L}^{-1}$  of silver chromate. The molar mass of silver chromate is  $331.8 \text{ g mol}^{-1}$ . The chromate ions react with hydrogen ions as shown below:



The molar concentration of chromate ions in a saturated solution of silver chromate in  $0.183 \text{ mol L}^{-1}$  silver nitrate is:

- (A)  $1.40 \times 10^{-12} \text{ mol L}^{-1}$   
(B)  $4.19 \times 10^{-11} \text{ mol L}^{-1}$   
(C)  $7.05 \times 10^{-3} \text{ mol L}^{-1}$   
(D)  $2.34 \times 10^{-2} \text{ mol L}^{-1}$

### Notes from the Marker



Solubility of  $\text{Ag}_2\text{CrO}_4 = 2.34 \times 10^{-2} \text{ g L}^{-1}$

$$\begin{aligned}\text{MM}(\text{Ag}_2\text{CrO}_4) &= (107.9 \times 2) + (52.00) + (16.00 \times 4) \\ &= 331.8 \text{ g mol}^{-1}\end{aligned}$$

$$\begin{aligned}[\text{Ag}_2\text{CrO}_4] &= \frac{2.34 \times 10^{-2} \text{ g L}^{-1}}{331.8 \text{ g mol}^{-1}} \\ &= 7.05 \times 10^{-3} \text{ mol L}^{-1}\end{aligned}$$

$$\begin{aligned}K_{\text{sp}} &= [\text{Ag}^+]^2[\text{CrO}_4^{2-}] \\ &= (2 \times 7.05 \times 10^{-3})^2 (7.05 \times 10^{-3}) \\ &= 1.40 \times 10^{-12} \text{ (3 s.f.)}\end{aligned}$$

To determine molar concentration of chromate ions in a saturated solution of silver chromate in  $0.183 \text{ mol L}^{-1}$  silver nitrate.

$$[\text{AgNO}_3] = 0.183 \text{ M}$$

$$[\text{Ag}^+] = [\text{NO}_3^-] = 0.183 \text{ M}$$

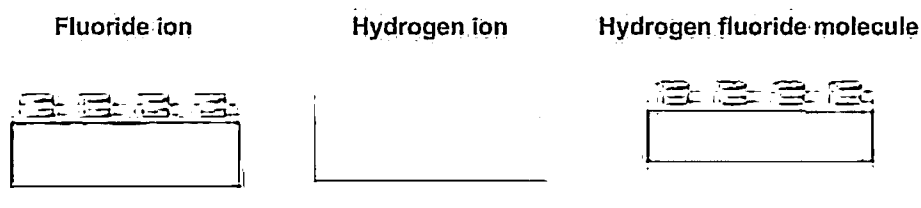
$$\text{Let } [\text{CrO}_4^{2-}] = s$$

$$1.4 \times 10^{-12} = (0.183)^2 \times s$$

$$s = 4.19 \times 10^{-11} \text{ mol L}^{-1}$$

Use the following information to answer questions 17–18.

A student was required to construct a model to help others visualise the composition of a solution of 0.1 M hydrofluoric acid. They were told this acid had a  $K_a$  value of  $6.7 \times 10^{-4}$ . The student had access to children's building blocks of various colours, like those shown in the image. The student counted out 25 dark blocks and 25 light blocks and created the following key.



The student used the  $K_a$  value provided to determine, via calculation, the appropriate number of dark and light blocks to join to make the number of 'HF molecules' in the model consistent with the  $K_a$  value for the acid.

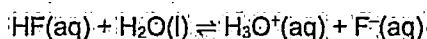
After joining the calculated number of blocks, they finally placed all single and joined blocks into a large beaker. This was presented as the final model.

**Question 17 (1 mark)**

How many separate dark blocks should the student have placed into the beaker to make the model show the correct percent ionisation for the acid?

- (A) 2
- (B) 6
- (C) 12
- (D) 23

**Notes from the Marker**



$$K_a = \frac{[\text{H}_3\text{O}^+]^2}{[\text{HF}]}$$

$$6.7 \times 10^{-4} = \frac{[\text{H}_3\text{O}^+]^2}{0.1}$$

$$[\text{H}_3\text{O}^+] = 0.00818535 \text{ M}$$

$$\begin{aligned} \text{Percent ionisation} &= \frac{0.00818535}{0.1} \times 100 \\ &= 8\% \\ 25 \text{ blocks of HF} \times 8\% &= 2 \text{ blocks.} \end{aligned}$$

**Question 18 (1 mark)**

Other students assessed the model and were asked to state some of its limitations based on the aim of the modelling exercise. Four statements made are shown below.

- I. The model should have included some different coloured blocks to represent the proportion of hydroxide ions also present in the solution.
- II. The building blocks used were too big to represent actual ions and molecules.
- III. It did not include the most common chemical present in the 0.1 M HF being modelled.
- IV. The true nature of the hydrogen ion in aqueous solution is not shown by the model.

Given the aim of the exercise, which of the above statements are significant limitations of the final model presented?

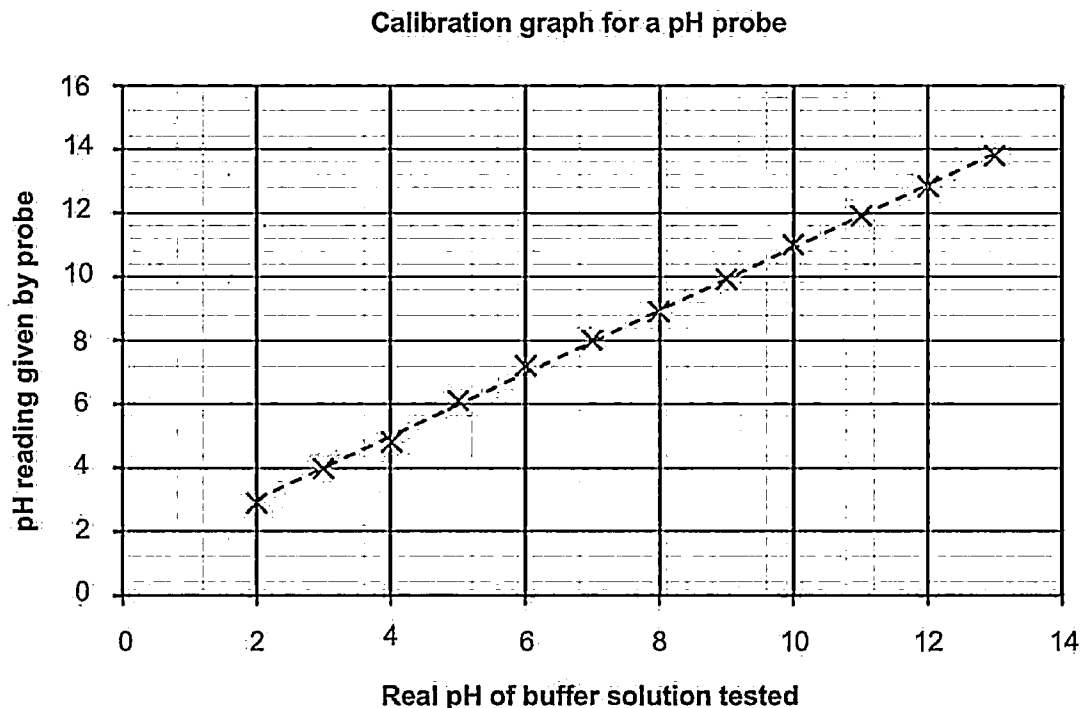
- (A) I and II
- (B) I and IV
- (C) II and III
- (D) III and IV

**Notes from the Marker**

- III is a limitation, as the model did not include chemicals such as  $\text{H}_2\text{O}$ , which is relevant in showing the composition of the acid solution as well as the behaviour of the acid.
- IV is a limitation, as the true nature of the hydrogen ion, being the hydronium ion ( $\text{H}_3\text{O}^+$ ) was not shown by the model.

**Question 19 (1 mark)**

A pH probe is used to measure the pH of several buffers of known pH values to determine if the meter required calibration. The graph below shows the pH readings given by the probe for each of the known (real) pH values of each buffer.



Which of the following statements is true, based on the data provided in the calibration graph shown?

- (A) Using the probe without calibration will affect the reliability of data it gathers.
- (B) If the probe were to be placed in a solution with  $[\text{OH}^-] = [\text{H}^+]$  it would read 6.
- (C) The equivalent point in a titration, as determined using this probe, will occur approximately one pH unit lower the actual equivalence point.
- (D) The  $[\text{H}^+]$  of a solution, as determined from its pH measured using this probe, will be 10 times lower than the actual  $[\text{H}^+]$  in the same solution.

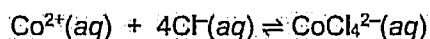
**Notes from the Marker**

A difference in a pH of 1 represents a 10-fold change in the acidity/basicity of a solution.

The pH readings given by the probe is always 1 pH higher than the real pH of the buffer solution tested. The pH probe therefore measures the  $[\text{H}^+]$  as 10 times lower than the actual  $[\text{H}^+]$  in the same solution.

**Question 20 (1 mark)**

A student performed an experiment to determine  $K_{eq}$  for the following reaction:



The student adds a 0.253 g sample of  $\text{CoCl}_2$  to a solution of 6.0 M  $\text{NaCl}$ . The value of  $K_{eq}$  is  $4.8 \times 10^{-4}$ .

Which of the following is the closest to the percentage of cobalt-containing species that is present as the blue  $\text{CoCl}_4^{2-}$  ion?

(A) 38%

(B) 46%

(C) 54%

(D) 62%

**Notes from the Marker**

$$K_{eq} = \frac{[\text{CoCl}_4^{2-}]}{[\text{Co}^{2+}][\text{Cl}^{-}]^4}$$

$$= 4.8 \times 10^{-4}$$

In the 6.0 M solution of  $\text{NaCl}$ ,  $[\text{Cl}^{-}] = 6.0 \text{ M}$

0.253 g of  $\text{CoCl}_2$  is added to this, introducing both  $\text{Co}^{2+}$  and  $\text{Cl}^{-}$ . To determine how much of each ion is added:

$$n(\text{CoCl}_2) = m/\text{MM}$$

$$= \frac{0.253}{58.93 + 2(35.45)}$$

$$= 0.001949 \text{ mol}$$

$$n(\text{Co}^{2+}) = 0.001949 \text{ mol}$$

$$n(\text{Cl}^{-}) = 0.001949 \times 2$$

$$= 0.003897 \text{ mol}$$

Hence, before equilibrium is established:

$$[\text{Co}^{2+}] = n/V$$

$$= \frac{0.001949}{0.500}$$

$$= 0.003897 \text{ M}$$

To find total  $[\text{Cl}^{-}]$  in solution initially:

$$n(\text{Cl}^{-})_{\text{total}} = n(\text{Cl}^{-})_{\text{initial}} + n(\text{Cl}^{-})_{\text{added}}$$

$$= (6.0 \times 0.500) + 0.003897$$

$$= 3.0 + 0.003897$$

$$= 3.0 \text{ mol}$$

$$[\text{Cl}^{-}] = n/V$$

$$= \frac{3.0}{0.500}$$

$$= 6.0 \text{ M}$$

Note that the amount of  $\text{Cl}^{-}$  added from the  $\text{CoCl}_2$  is negligible.

Use an ICE table to determine how much of  $\text{Co}^{2+}$  initially becomes  $\text{CoCl}_4^{2-}$ :

|             | $[\text{Co}^{2+}]$ | $[\text{Cl}^{-}]$ | $[\text{CoCl}_4^{2-}]$ |
|-------------|--------------------|-------------------|------------------------|
| Initial     | 0.003897           | 6.0               | 0.0                    |
| Change      | $-x$               | $-4x$             | $+x$                   |
| Equilibrium | $0.003897 - x$     | $6.0 - 4x$        | $x$                    |

Since  $x \ll 6.0$ ,  $[\text{Cl}^{-}] \approx 6.0 \text{ M}$  (assuming  $x$  is negligible compared to 6.0 M).

$$K_{eq} = \frac{[\text{CoCl}_4^{2-}]}{[\text{Co}^{2+}][\text{Cl}^{-}]^4}$$

$$4.8 \times 10^{-4} = \frac{[\text{CoCl}_4^{2-}]}{[\text{Co}^{2+}] \times 6.0^4}$$

$$\frac{[\text{CoCl}_4^{2-}]}{[\text{Co}^{2+}]} = 4.8 \times 10^{-4} \times 6.0^4$$

$$= 0.62$$

$$[\text{CoCl}_4^{2-}] = 0.62 \times [\text{Co}^{2+}]$$

This means that for every 100  $\text{CoCl}_4^{2-}$ , we have only ~62  $\text{Co}^{2+}$  ions. Thus, out of a total of 162 cobalt-containing species, 62 of these would be  $\text{Co}^{2+}$  ions.

The percentage would therefore be:

$$\frac{62}{162} \times 100 = 38\%$$



# 2022

## NBHS Year 12 Chemistry Trial Examination

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

**Sample**     $2 + 4 =$     (A) 2    (B) 6    (C) 8    (D) 9

A ☐    B ☒    C ☐    D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒    B ☒    C ☐    D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒    B ☒ <sup>correct</sup>    C ☐    D ☐

- |    |                                    |                                    |                                    |                                    |
|----|------------------------------------|------------------------------------|------------------------------------|------------------------------------|
| 1  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 2  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 3  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 4  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 5  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 6  | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 7  | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 8  | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 9  | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input type="radio"/>            |
| 10 | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 11 | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 12 | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 13 | A <input type="radio"/>            | B <input type="radio"/>            | C <input checked="" type="radio"/> | D <input checked="" type="radio"/> |
| 14 | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 15 | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 16 | A <input type="radio"/>            | B <input checked="" type="radio"/> | C <input type="radio"/>            | D <input type="radio"/>            |
| 17 | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |
| 18 | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 19 | A <input type="radio"/>            | B <input type="radio"/>            | C <input type="radio"/>            | D <input checked="" type="radio"/> |
| 20 | A <input checked="" type="radio"/> | B <input type="radio"/>            | C <input type="radio"/>            | D <input type="radio"/>            |

# Normanhurst Boys High School

## 2023 Year 12 Chemistry Trial Examination

### Marking Responses, Sample Answers & Notes from the Marker



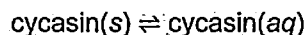
#### Section 2A (10 marks)

#### Question 1

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <p>Identifies that macrozamin and cycasin can form bonds with water, with reference to the relevant structural group (hydroxyl).</p> <p>AND</p> <p>States the equilibrium reaction that can occur with the toxins in water (either described in words or written as an equation).</p> <p>AND</p> <p>Justifies the observation to remove toxins by:</p> <ul style="list-style-type: none"> <li>Describing the difference between running water (open system) and still water (closed system, equilibrium forms) OR states why still water would not be efficient by relating to equilibrium.</li> <li>Relating to the efficiency of running water in removing toxins through LCP.</li> </ul> <p><b>Response shows a good understanding of principles involving dissolving and equilibrium.</b></p>                                                                                                                                                                                                                 | 3     |
| <p>Identifies that macrozamin and cycasin can form bonds with water.</p> <p>AND</p> <p>Justifies the observation to remove toxins by:</p> <ul style="list-style-type: none"> <li>Relating to the efficiency of running water in removing toxins through LCP.</li> </ul> <p>OR:</p> <p>Justifies the observation to remove toxins by:</p> <ul style="list-style-type: none"> <li>Describing the difference between running water (open system, no equilibrium) and still water (closed system, equilibrium forms)</li> <li>Relating to the efficiency of running water in removing toxins through LCP.</li> </ul> <p>OR:</p> <p>States the equilibrium reaction that can occur with the toxins in water.</p> <p>AND</p> <p>Justifies the observation to remove toxins by:</p> <ul style="list-style-type: none"> <li>Relating to the efficiency of running water in removing toxins through LCP.</li> </ul> <p><b>Response shows a sound understanding of principles involving dissolving and equilibrium.</b></p> | 2     |
| <p>Provides some relevant information, such as ONE of the following:</p> <ul style="list-style-type: none"> <li>Identifies that the toxins can form hydrogen bonds with water.</li> <li>States the equilibrium reaction that can occur with the toxins in water.</li> <li>Provides a reason for why running water is more efficient than still water</li> <li>Stated a reason for the crushed powder – to increase surface area to be more efficient at dissolving.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1     |

## Sample Response

The toxins macrozamin and cycasin have multiple **polar hydroxyl groups**, which allow for multiple **hydrogen bonding to occur with water**. This allows the toxins to **dissolve** in water, forming an equilibrium, for example:



Still water would form an equilibrium as it is a **closed system**, while running water is an **open system** so the **dissolved toxins do not reach equilibrium**. As a result, running water is more efficient than still water in removing the toxins, as it allows the dissolved toxins to be flushed out. It ensures that the equilibrium is shifted to the right, as per Le Chatelier's principle, as the dissolved toxins are constantly removed. This results in more toxins being dissolved and continually flushed away, eventually removing the toxins.

## Notes from the Marker

- Better responses:
  - showed a good understanding in the concept of dissolving and specified the intermolecular forces formed with the toxins and water.
  - were clear when relating the above observation to equilibrium principles, specifying how running water would affect the equilibrium.
- Weaker responses:
  - did not show a good understanding in the concept of dissolving, stating that a chemical reaction is occurring instead of the intermolecular forces formed with water.
  - did not address the comparison of running water to still water. These responses need to be explicit to show why running water is **more** efficient.

## Student sample (student scored 3/3)

With reference to structural formulae and relevant chemical principles, justify the above observation used to remove the toxins.

<sup>and macrozamin</sup>  
Cycasin can form hydrogen bonds with water, allowing it to dissolve, due to the high presence of polar O-H bonds.  
 $\text{Cycasin(s)} \rightleftharpoons \text{Cycasin(aq)}$  and  $\text{macrozamin(s)} \rightleftharpoons \text{macrozamin(aq)}$   
When placed in still water, the toxins shift right to establish equilibrium, thus there is a lower amount of cycasin left in the solid, however cycasin is still present in the solid when equilibrium is formed. Placing the bag under running water shifts the equilibrium right. As the system is open, the toxins are unable to form solid cycasin as it is removed from the system by the running water, thus this process will eventually reach out all the solid cycasin until none remains; this is more efficient than the still water method.

## Question 2

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <p>Correctly identifies group A and B.<br/> AND<br/> Identifies the <b>THREE</b> trends in the graph.<br/> AND<br/> Explicitly relates the <b>structure</b> of alkanes and alkane nitriles to their <b>intermolecular forces</b> (i.e. non-polar, polar, specific bonds).<br/> AND<br/> Explains <b>EACH</b> of the trends by relating to how intermolecular forces influence boiling point.<br/> AND<br/> Relates the strength of the intermolecular forces to boiling point.<br/> AND<br/> Explains what effects dispersion forces have on boiling point (when increasing molar mass).<br/> AND<br/> Relates the effect of different contributions of the intermolecular forces to the trends shown.</p> <p><b>Response is coherent, uses correct terminology, links to the stimulus provided and explanations demonstrate a deep understanding of how intermolecular forces influence boiling point.</b></p> | 5     |
| <p>Correctly identifies group A and B.<br/> AND<br/> Identifies at least <b>TWO–THREE</b> trends in the graph.<br/> AND<br/> Relates the structure of alkanes and alkane nitriles to their intermolecular forces.<br/> AND<br/> Explains at least <b>TWO</b> of the trends by relating to how intermolecular forces influence boiling point.<br/> AND<br/> Relates the strength of the intermolecular forces to boiling point.<br/> AND/OR<br/> Explains what effects dispersion forces have on boiling point (when increasing molar mass).<br/> AND/OR<br/> Relates the effect of different contributions of the intermolecular forces to the trends shown.</p> <p><b>Response uses correct terminology and explanations demonstrate a good understanding of how intermolecular forces influence boiling point.</b></p>                                                                                        | 4     |
| <p>Correctly identifies group A and B.<br/> AND<br/> Identifies at least <b>ONE–TWO</b> trends in the graph.<br/> AND<br/> Relates the relevant molecules to their intermolecular forces.<br/> AND<br/> <b>Sound</b> explanation of at least <b>ONE</b> of the trends by relating to how intermolecular forces influence boiling point by using <b>correct terminology to identify the intermolecular forces that caused the trend.</b><br/> AND/OR<br/> Relates the strength of the intermolecular forces to boiling point.</p>                                                                                                                                                                                                                                                                                                                                                                                | 3     |
| <p>Correctly identifies group A and B.<br/> AND<br/> Identifies at least <b>ONE</b> trend from the graph.<br/> AND<br/> Relates the structure of the molecules to their intermolecular forces.<br/> AND/OR<br/> Relates the strength of the intermolecular forces to boiling point.<br/> AND<br/> <b>Responses shows a limited understanding of the intermolecular forces that caused the trend.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2     |

Provides some relevant information, such as ONE of the following:

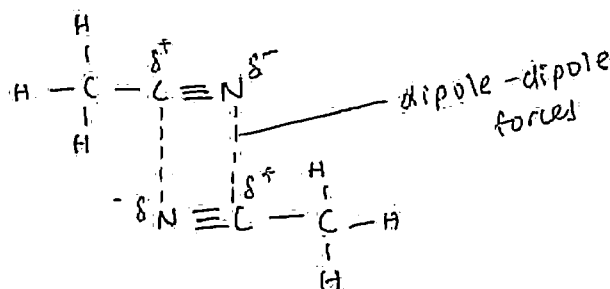
1

- Identifies at least ONE trend from the graph AND outlines a reason for the trend.
- Identifies the intermolecular forces present in alkanes and alkane nitriles.

### Sample Response

A shows the data points for alkane nitriles and B shows the data points for alkanes.

Alkane nitriles would have a higher boiling point than alkanes because they can form **dipole-dipole forces** between molecules due to the **C≡N bond**, as shown. There is also greater **electron density** at the triple bond, making this portion of the molecule **polar**.



Alkanes however, only form **dispersion forces** between molecules, as the molecules are **non-polar**. Since **dipole-dipole forces are stronger than dispersion forces**, alkane nitrile molecules **require more energy to break these intermolecular forces**, resulting in a **higher boiling point**.

As the chain length increases for both alkanes and alkane nitriles, the boiling point for both groups also increase. This is due to **stronger dispersion forces** between the carbon chain length, due to a **greater electron clouds** and greater surface area between molecules, therefore resulting in a higher boiling point.

As chain length increases, the difference in boiling points between the two groups also decreases, as shown with the 2 carbon chain length alkane and alkane nitrile with a difference in boiling point of around 170°C, while the 6 carbon chain length molecules have a difference in boiling point of around 100°C between the two groups. This is due to the **dipole-dipole forces** between alkane nitrile molecules having a smaller contribution to the overall intermolecular forces and is **outweighed by the contribution of dispersion forces** from the longer carbon chains.

### Notes from the Marker

- Better responses:
  - were explicit and provided specific details, such as linking the strength of intermolecular forces to boiling point and related the two structures to its intermolecular forces.
  - made specific references to the stimulus by providing specific values.
  - were coherent and showed a deep understanding in their explanations of the trends and application of their knowledge.
- Weaker responses:
  - did not **explicitly** relate the strength of intermolecular forces to boiling point (i.e. linking back to the energy required to break those forces).
  - incorrectly showed where the intermolecular forces are between alkane nitriles; where the dipole-dipole forces were incorrectly shown as N bonded to the H of the non-polar part of the alkane nitrile.
  - incorrectly stated that alkane nitriles can form hydrogen bonds instead of dipole-dipole forces.
  - provided incorrect information, such as describing "rate" when interpreting the graph.

## Student sample 1

5

Identify the label of A and B as either alkanes or alkane nitriles and explain the patterns of the boiling points shown in the graph.

~~Trend 1:~~ A: alkane nitriles B: alkanes

Trend 1: For the same number of carbons, the boiling point of

alkane nitriles is significantly higher than that of alkanes.

This is because alkane nitriles contain a polar  $\text{C}\equiv\text{N}$  group, which allows the molecule to form strong dipole-dipole forces between

~~additionally~~ as well as dispersion forces. As a result, more energy is required to separate molecules so B.P. is higher seen in graph where

for 2 carbons,  $\text{BP}(\text{alkanes}) = -90^\circ\text{C}$  which is less than  $\text{BP}(\text{alkane nitriles}) = 80^\circ\text{C}$ .

Trend 2: As the number of carbons increases, the boiling point for both molecules increases. This is because the addition of more atoms,

introduces more electrons which ~~increase~~ <sup>makes</sup> the electron cloud more polarizable, allowing stronger dipole-dispersion forces to form which require more energy to separate. It thus has a higher boiling point.

Seen in graph where  $\text{BP}(\text{alkanes})$  increases from  $-90^\circ\text{C}$  to  $70^\circ\text{C}$  from 2 to 6 atoms and  $\text{BP}(\text{alkane nitriles})$  increases from  $80^\circ\text{C}$  to  $165^\circ\text{C}$  from 2 to 6 atoms.

Trend 3 As the number of atoms increases, the difference in boiling point between two molecules decreases.

This is because, adding more carbons increases strength of dispersion forces only. The polar end of the molecule in alkane nitriles remains the same, so the dipole-dipole forces contribute less to overall intermolecular forces, which means as number of C atoms  $\uparrow$  from 2-6, difference in BP decreases from  $170^\circ\text{C}$  to  $95^\circ\text{C}$  because the difference in energy required to separate molecules decreases.

## Student sample 2

5

Identify the label of A and B as either alkanes or alkane nitriles and explain the patterns of the boiling points shown in the graph.

Alkanes are non-polar and bond with weak dispersion forces, while nitriles have polar  $C \equiv N$  bonds, which allow them to form stronger dipole-dipole forces ~~than~~ <sup>and</sup> dispersion interactions.

Propane  $-40^\circ C$  BP      propanenitrile  $118^\circ C$  BP

$$\begin{array}{c}
 H & H & H \\
 | & | & | \\
 H-C-C-C-H \\
 | & | & | \\
 H & H & H
 \end{array}
 \quad
 \begin{array}{c}
 H & H & \\
 | & | & \\
 H-C-C-C \equiv N \\
 | & | & | \\
 H & H & N \equiv C-K
 \end{array}$$

$\delta^-$   $\delta^+$  (capable of dipole-dipole)

1. Hence, the intermolecular forces between nitriles require more thermal energy to break compared to alkanes of the same length, resulting in a higher BP.
2. As chain length increases, the number of electrons in both nitriles and alkanes increases, resulting in stronger dispersion forces in both. This means as chain length increases, both alkanes and nitriles see an increase in BP (5 carbon nitriles have a BP of  $146^\circ C$  compared to 2 carbon nitriles with a BP of  $28^\circ C$ ).
3. As chain length increases, the strength of dispersion forces in nitriles increases while the strength of the dipole-dipole forces remains the same. Hence, the difference in boiling point compared to alkanes decreases due to the smaller proportional increase in intermolecular strength. (as shown in graph) <sup>(compared to alkanes)</sup> The difference in boiling point at  $C=5$  is  $105^\circ$ , less than the difference at  $C=2$  ( $17^\circ C$ ).

### Question 3(a)

| Marking Criteria                                            | Marks |
|-------------------------------------------------------------|-------|
| Provides an appropriate reason for the observation of soot. | 1     |

### Sample Response

Incomplete combustion may have occurred from the burner flame, which produces soot.

#### Notes from the Marker

- Better responses were specific in being able to identify that incomplete combustion resulted in the formation of soot.
- Weaker responses:
  - restated the information from the question, such as stating that soot was formed.
  - were not specific in the reason for soot, instead stating "insufficient oxygen".

### Question 3(b)

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Correctly calculates the moles of butan-1-ol.<br>AND<br>Relevant steps to calculate the total quantity of heat released.<br>AND<br>Relevant steps to calculate the quantity of heat measured after heat loss.<br>AND<br>Correctly calculates the specific heat capacity of the liquid, including units, to appropriate significant figures.<br><br><b>Response shows a thorough understanding of heat of combustion.</b> | 4     |
| All correct steps with any ONE error.                                                                                                                                                                                                                                                                                                                                                                                    | 3     |
| Correct steps with incorrect answer and multiple errors.                                                                                                                                                                                                                                                                                                                                                                 | 2     |
| Any ONE step correct only.                                                                                                                                                                                                                                                                                                                                                                                               | 1     |

### Sample Response

$$n(\text{C}_4\text{H}_9\text{OH}) = \frac{m}{M_r}$$

$$= \frac{0.375}{4(12.01) + 10(1.008) + 16.00}$$

$$= 0.00505936319 \text{ mol}$$

$$\Delta H = 2676 \text{ kJ mol}^{-1}$$

$$\Delta H = \frac{q}{n}$$

$$q_{\text{total}} = \Delta H \times n$$

$$= 2676 \times 0.00505936319$$

$$= 13.53885591 \text{ kJ}$$

$$= 13538.86 \text{ J}$$

35% heat loss, so 65% heat absorbed:

$$q_{\text{measured}} = 13538.86 \times 0.65$$

$$= 8800.256 \text{ J}$$

$$q = mC\Delta T$$

$$8800.256 = 150 \times C \times (40.0 - 25.0)$$

$$C = 3.91 \text{ J g}^{-1} \text{ K}^{-1}$$

## Notes from the Marker

- Better responses:
  - showed clear working out by showing each step and calculating their answers at each step.
  - were able to interpret the stimulus correctly and determined the change in temperature using the graph and applied the heat loss correctly.
- Weaker responses:
  - did not apply the percentage of heat loss correctly, where they did not multiply the heat by 65.0%, which was the heat measured (since the question stated that 35.0% of heat was lost to the environment).
  - showed an incorrect conceptual understanding of the heat loss. To account for the heat loss, it should be applied to either  $q$  or  $\Delta H$ , however, some responses applied it to the temperature change, which is conceptually and practically incorrect.
  - did not show clear working out and took many shortcuts. These responses resulted in incorrect answers and not enough steps shown in the working out. Each step should be clearly shown and calculated.
  - rounded up numbers on the periodic table when calculating molar mass, which resulted in an incorrect final answer.

## Student sample

- (a) The student noted that there was some soot on the base of the beaker.

1

Outline a reason for this observation.

~~for~~ incomplete combustion of butan-1-ol occurred due to limited oxygen supply thus forming soot on the beaker.

- (b) Based on prior testing, it was determined that 35.0% of the heat produced by the combustion of butan-1-ol is lost to the surroundings.

4

Using the information provided, calculate the specific heat capacity of the liquid, given that the heat of combustion of butan-1-ol is  $2676 \text{ kJ mol}^{-1}$ .

$$\begin{aligned}
 n(\text{C}_4\text{H}_9\text{OH}) &= \frac{m}{M_r} \\
 &= \frac{0.375}{4(12.01) + 10(1.008) + 16.00} \\
 &= 0.005059 \text{ mol} \\
 q &= \Delta H n \\
 &= 2676 \times 0.005059 \text{ mol} \\
 &= 13538.5591 \text{ kJ} \\
 &= 13538.5591 \text{ J}
 \end{aligned}$$

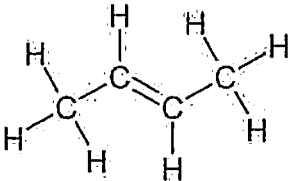
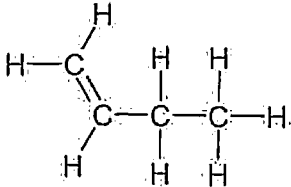
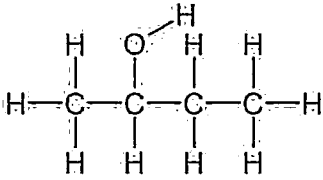
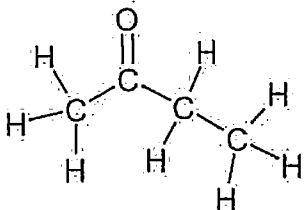
$$\begin{aligned}
 q_{\text{actual}} &= 0.65 \times 13538.5591 \\
 &= 8800.256341 \text{ J} \\
 q &= mc\Delta T \\
 8800.256341 &= 50 \times c \times (40 - 25) \\
 c &= \frac{8800.256341}{150 \times (40 - 25)} \\
 &= 3.91 \text{ J g}^{-1} \text{ K}^{-1}
 \end{aligned}$$

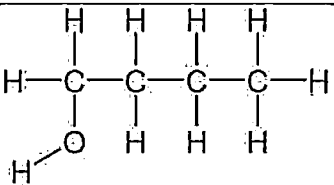
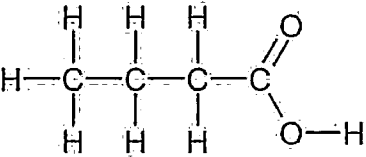
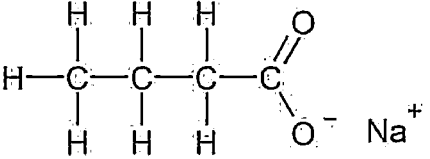
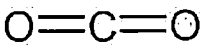
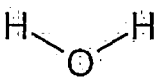
#### Question 4

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <p>All correct structures and correct IUPAC names for compounds A, B, C, D, E, F, G, H and I.<br/>AND<br/>Correct justifications for all compounds A to I, including:</p> <ul style="list-style-type: none"> <li>• A and B being isomers</li> <li>• how compound A is determined to be symmetrical with a double bond.</li> <li>• hydration reaction to produce alcohols</li> <li>• oxidation of alcohol shows it is either primary or secondary</li> <li>• the reaction with <math>\text{NaHCO}_3</math> determines the presence of the acid F</li> <li>• acid carbonate reaction to produce G, H and I</li> </ul> <p>AND<br/>Key terms used in justification.<br/>AND<br/>Makes <b>explicit references to most of the information</b> provided in the flowchart.</p>                                                                                                                                 | 7     |
| <p>All correct structures and correct IUPAC names for compounds A, B, C, D, E, F, G, H and I.<br/>AND<br/>Correct justifications for all compounds A to I, including:</p> <ul style="list-style-type: none"> <li>• how compound A is determined to be symmetrical with a double bond.</li> <li>• hydration reaction to produce alcohols</li> <li>• oxidation of alcohol shows it is either primary or secondary</li> <li>• the reaction with <math>\text{NaHCO}_3</math> determines the presence of the acid F</li> </ul> <p>AND<br/>Key terms used in justification.<br/>AND<br/>Makes <b>limited references</b> to the information provided in the flowchart.</p> <p><b>All of the above but contains ONE type of error</b> (such as incorrect locants used when naming, incorrect commas used when naming, a few key terms missing) OR all of the above criteria but 1–2 minor types of errors.</p> | 6     |
| <p>All of the above but contains 2-3 types of errors.<br/>AND/OR<br/>All IUPAC names and structures mentioned but key terms missing in justification.<br/>AND/OR<br/>All of the above but missing some names.<br/>OR<br/>1 missing structure but correct IUPAC name and correct justification for all other compounds.<br/>OR<br/>No structures but correct IUPAC names and correct justifications for all compounds.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 5     |
| <p>Response contains 2-4 errors.<br/>AND/OR<br/>Correct structures with no names and some justifications.<br/>OR<br/>1-5 incorrect or missing structures or names and justifications incomplete.<br/>OR<br/>4 correct structures with names and appropriate justifications</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 4     |
| <p>Response contains at least 5 errors.<br/>OR<br/>Some structures or names are incorrect but some correct justifications.<br/>OR<br/>1-6 incorrect or missing structures or names, with very limited justifications.<br/>OR</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3     |

|                                                                                                                                                                                                                                                                                                                                                                                           |   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 3 correct structures with names and appropriate justifications                                                                                                                                                                                                                                                                                                                            |   |
| <p>Response contains 5 or more errors.</p> <p>OR</p> <p>Most structures or names are incorrect and limited justifications.</p> <p>OR</p> <p>Structures are incorrect AND/OR have some correct IUPAC names AND shows some understanding in following the reactions in the flowchart and no justification (shows <u>understanding</u> in at least two reactions/pathways in flowchart).</p> | 2 |
| <p>Some correct/relevant information and shows limited understanding in following the reactions in the flowchart, such as ONE of the following:</p> <ul style="list-style-type: none"> <li>Shows understanding in ONE of the reactions.</li> <li>Identifies the structure of any ONE compound.</li> <li>Identifies a structure of the <math>C_4H_8</math> compound.</li> </ul>            | 1 |

### Sample Response

| Compound | Structure and Name                                                                                     | Justification                                                                                                                                                                                                                                                                       |
|----------|--------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A        | <p>but-2-ene</p>     | Compound A undergoes <b>hydration</b> with <b>dilute <math>H_2SO_4</math></b> to produce <b>only one alcohol product</b> . It is therefore a <b>symmetrical</b> , unsaturated hydrocarbon with the double bond at the second carbon, with the <b>formula <math>C_4H_8</math></b> .  |
| B        | <p>but-1-ene</p>    | The hydration of compound B with dilute $H_2SO_4$ produces <b>two possible alcohol products</b> , so it is <b>unsymmetrical</b> , with the double bond on first carbon, since one of the product, C, is also produced by A. It is therefore a straight chained <b>isomer of A</b> . |
| C        | <p>butan-2-ol</p>   | Compound C can <b>oxidise</b> when <b>acidified <math>KMnO_4</math></b> is added and since the double bond is at the second carbon, it is a <b>secondary alcohol</b> .                                                                                                              |
| D        | <p>butan-2-one</p>  | Compound D is the <b>product of the oxidation reaction</b> of butan-2-ol. It also <b>does not react</b> with $NaHCO_3$ so it is a <b>ketone</b> , not a <b>carboxylic acid</b> .                                                                                                    |
| E        | <p>butan-1-ol</p>                                                                                      | Compound E is the other product (major product) of the <b>hydration with dilute <math>H_2SO_4</math></b> reaction of compound B.                                                                                                                                                    |

|   |                                                                                                           |                                                                                                                                                               |
|---|-----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   |                          |                                                                                                                                                               |
| F | <p>butanoic acid</p>     | Since compound E is the <b>primary alcohol</b> , it <b>oxidises</b> to produce a <b>carboxylic acid</b> (compound F) when acidified $\text{KMnO}_4$ is added. |
| G | <p>sodium butanoate</p>  | From the <b>acid-carbonate reaction</b> between compound F and $\text{NaHCO}_3$ , a salt, carbon dioxide and water are produced.                              |
| H | <p>carbon dioxide</p>    |                                                                                                                                                               |
| I | <p>water</p>            |                                                                                                                                                               |

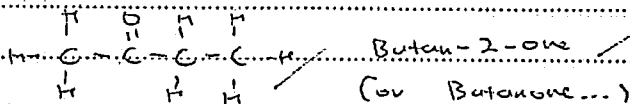
### Notes from the Marker

- Better responses:
  - made specific references to the stimulus information, included key terms and provided sufficient detail to justify their structures.
  - correctly provided the IUPAC name for each structure and correctly drew all the structures.
- Weaker responses:
  - needed more specific references to the stimulus and required more key terms in their justification. They also needed more detail to justify their compounds.
  - did not state the IUPAC names of all of the compounds or incorrectly stated the IUPAC names.
  - incorrectly drew their structural formulae and did not expand out the bonds (such as between O-H), did not include all the H's on the hydrocarbon chain, or incorrectly showed a 5 bonded carbon.
  - incorrectly drew the structure of carbon dioxide and sodium butanoate.

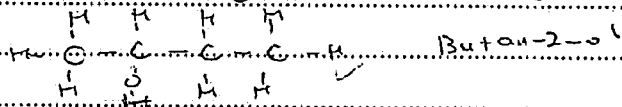
## Student sample

Draw the structural formulae and state the IUPAC names of compounds A to I, justifying your compounds with reference to the information provided in the flow chart.

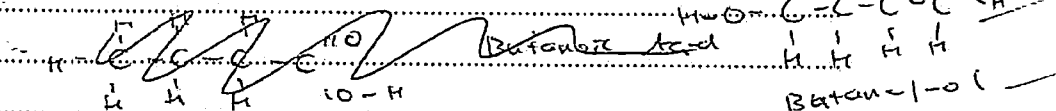
- (P) does not react with  $\text{NaHCO}_3$  but does oxidise to an alcohol  
C. This means that D could only possibly be a ketone with 4 carbon atoms



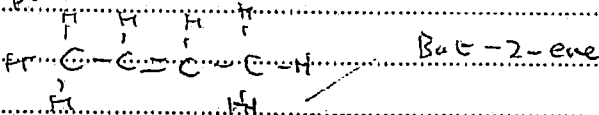
- (D) oxidised to give P, which is a ketone so C must be a secondary alcohol with 4 carbon atoms



- (E) must be a primary alcohol with 4 carbon atoms as it oxidises to form F, which is a carboxylic acid as it reacts with  $\text{NaHCO}_3$

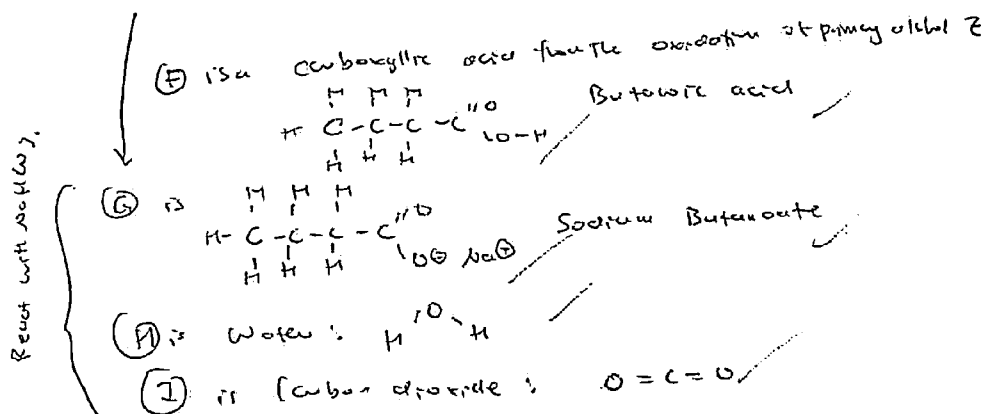
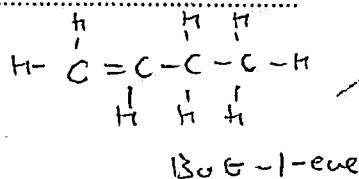


- (F) must be an alkene (as 4H is same as C<sub>4</sub>H<sub>8</sub> for alkenes) with the double bond on the secondary carbon as it undergoes hydration (H<sub>2</sub>O with H<sup>+</sup> catalyst) to form only one product



- (B) Hence must be the carbon on the 1st carbon of the alkene C it forms 2 products, major and minor depending on how the -OH from the water is added

END OF SECTION 2A



# Normanhurst Boys High School

## 2023 Year 12 Chemistry Trial Examination

### Marking Responses, Sample Answers & Notes from the Marker

S2B

#### Section 2C (40 marks) Written Response

##### Question 1(a)

| Marking Criteria                                                                                                | Mark |
|-----------------------------------------------------------------------------------------------------------------|------|
| • All relevant steps shown clearly AND correct answer to appropriate decimal places (i.e. to 2 decimal places). | 1    |
| • Correct answer with no working OR non attempts.                                                               | 0    |

##### Sample Response

$$\frac{[A^-]}{[HA]} = \frac{\frac{1}{2}}{\frac{1}{2}} = 1 \quad pK_a = -\log_{10}(3.0 \times 10^{-10}) = 9.52$$
$$pH = 9.52 + \log_{10} 1 = 9.52 \quad (2 \text{ d.p.})$$

##### Marker Feedback:

The number of decimal places in pH should equal to the number of significant figures in concentration.

##### Question 1(b)

| Marking Criteria                                                                      | Marks |
|---------------------------------------------------------------------------------------|-------|
| • Identifies range for colour change AND links choice of indicator to the given range | 2     |
| • Correct answer with no working OR non attempts                                      | 1     |

##### Sample Response

Range for change of colour =  $pK_a \pm 1.00 = 9.52 \pm 1$   
Range: 8.52 – 10.52 — ①

pH of 9.30 falls in the above range  
So HPH is a suitable indicator as its colour change falls within 8.52 – 10.52. — ①

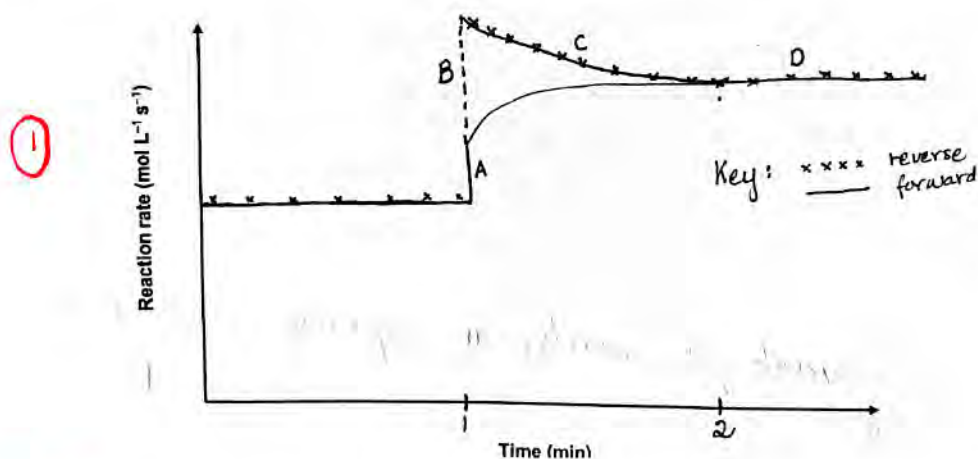
##### Marker Feedback:

- You cannot score for repeating statements from part (a).
- Question was well answered. Most students understand the use/choice of indicators. Some students lost marks for not showing all working.

Question 2

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Draws a correct graph, neat, using most of the given space with appropriate key for reverse and forward reaction pathway.<br/>AND</li> <li>• Explains parts A, B C AND D using both collision theory and LCP where most appropriate<br/>AND</li> <li>• Explanation written using all key terminology<br/>AND</li> <li>• Explanation is succinct and maintains chronological manner</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>• All the above EXCEPT One key feature missing<br/>OR</li> <li>• Section A or B not explained AND uses collision theory AND/OR LCP in detail and draws correct graph</li> </ul>                                                                                                                                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>• Incorrect graph AND Section B NOT explained<br/>AND</li> <li>• A, C, D outlined using collision theory AND/OR LCP<br/>OR</li> <li>• Correct graph and any two sections not explained in detail</li> </ul>                                                                                                                                                                                     | 2     |
| <ul style="list-style-type: none"> <li>• Correct graph<br/>OR</li> <li>• Any one section explained correctly using Collision Theory AND/OR LCP<br/>OR</li> <li>• All sections outlined with limited detail</li> </ul>                                                                                                                                                                                                                  | 1     |

## Sample Response



- ① A expl<sup>n</sup> Increase in T, increases reaction rate (A) for the whole reaction.
- LCP/LCP This is due to excess heat energy absorbed by the system. This in turn increases KE, hence greater collisions between particles. This increases successful collisions for all molecules therefore both reactant formation (reverse rate) and product formation (forward rate) increases. The endothermic reverse reaction absorbs heat energy to a greater extent (B) than the exothermic forward reaction. Thus  $[\text{N}_2, \text{H}_2]$  will increase more than  $[\text{NH}_3]$ . The system will counter this by shifting equilibrium to the right in an attempt to reduce the  $[\text{N}_2, \text{H}_2]$ . Thus the rate of forward reaction will increase while the rate of reverse will slow down and at  $t=2 \text{ min}$  equilibrium will be re-established. (LCP)
- ① B & C expl<sup>n</sup>
- LCP

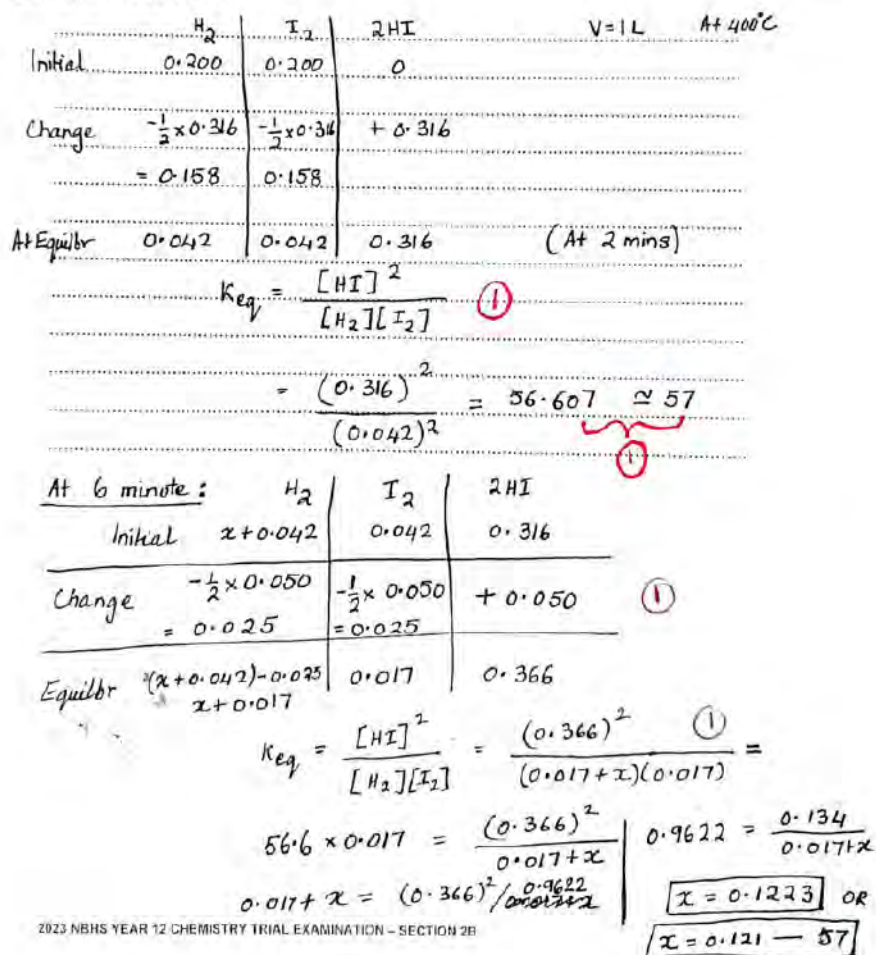
### Marker's Feedback:

- Question was well answered. Most students explained well. Graphs require further practice in neatness, use of full space, using key and labelling.
- Labelled sections from the graph should be used to support explanation.
- Some students failed to explain section B. There is NO requirement to state LCP. Collision theory and LCP should be used to explain only. Therefore, key terminologies are critical to score full marks.

## Question 3

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math> AND</li> <li>Shows all relevant values for ICE table 2 at 6 minutes AND</li> <li>Sets up correct <math>K_{eq}</math> expression for table 2 AND</li> <li>Solves algebraic equation and calculates the moles of hydrogen added (shown as x). <math>x = 0.1223</math> OR <math>0.121</math>.</li> </ul> | 5     |
| <ul style="list-style-type: none"> <li>All relevant steps correct with any ONE calculation error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     | 4     |
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math> AND</li> <li>Error in ICE table 2 AND error in algebraic calculation leading to two significant errors AND</li> <li>Incorrect value for x.</li> </ul>                                                                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math>.</li> </ul>                                                                                                                                                                                                                                                                                           | 2     |
| <ul style="list-style-type: none"> <li>Any one step of calculation or expression correct.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | 1     |

## Sample Response



## Question 4

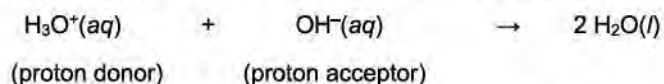
| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Uses B-L Theory labels such as proton donor (BL acid) and BL base (proton acceptor) (acid-base chemistry) to explain reaction between strong acids and strong bases<br/>AND</li> <li>• Shows equation for neutralization as <math>\text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l})</math> only for all above neutralisation<br/>AND</li> <li>• Justifies above equation to support <math>\Delta H = -57.6 \text{ kJ mol}^{-1}</math> for all strong acid/base neutralisations<br/>AND</li> <li>• Identifies <math>\text{NH}_3</math> as weak base and explains its ionization in water using B-L theory<br/>AND</li> <li>• Shows <b>ALL</b> relevant equations OR writes all word equations<br/>AND</li> <li>• Explains HCl reacting directly with ammonia instead of <math>\text{OH}^-</math><br/>OR</li> <li>• Explains correctly differently why <math>\text{NH}_3</math> with HCl leads to a different <math>\Delta H</math> value</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>• Outlines or explains B-L Theory labels to explain reaction between strong acids and strong bases<br/>AND</li> <li>• Outlines/explains the reaction between strong acids such as HCl and <math>\text{NH}_3</math><br/>AND</li> <li>• Uses one/two relevant equations only related to any one of the above</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>• Provides detailed justification for strong acid/base neutralisation with appropriate equation as <math>\text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l})</math><br/>AND/OR</li> <li>• Provides correct equation for ammonia and HCl as <math>\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4^+ + \text{Cl}^-</math> but no explanation for ammonia and HCl is provided<br/>OR</li> <li>• Incorrect justification ammonia and HCl is provided</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
| <ul style="list-style-type: none"> <li>• Explains any one of the above concepts well<br/>OR</li> <li>• Outlines both concepts with NO chemical equation/incorrect equation with multiple errors in understanding<br/>OR</li> <li>• No explanation but one correct equation only<br/>OR</li> <li>• Outlines both concepts but no chemical equation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1     |

## Sample Response

### Explaining same enthalpy of neutralisation for strong acids with strong bases:

When solutions of a strong acid (such as HCl/HNO<sub>3</sub>) are mixed with a strong base (such as NaOH /KOH), H<sub>3</sub>O<sup>+</sup> and OH<sup>-</sup> ions are undergoing neutralisation due to 100% ionisation **only**, all other ions are such as Cl<sup>-</sup>, Na<sup>+</sup>, NO<sub>3</sub><sup>-</sup> are spectators.

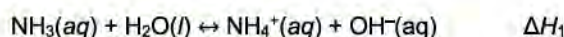
Hence, strong acid/base neutralisation reaction is always as:



The value for the enthalpy of neutralisation when NaOH reacts with HCl and between HNO<sub>3</sub> and KOH are therefore the same and are equal to -57.6 kJ mol<sup>-1</sup>.

### Explaining the difference in enthalpy of neutralisation for strong acid HCl with weak base ammonia NH<sub>3</sub>:

NH<sub>3</sub> is a **weak base (proton acceptor)**. The reaction equation for the reaction that occurs when NH<sub>3</sub> is dissolved in water is shown below:



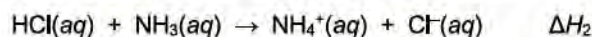
Therefore, a solution of NH<sub>3</sub> would **only contain a small amount of OH<sup>-</sup>** and NH<sub>4</sub><sup>+</sup> ions, only occurs to a small extent. There would be a significant amount of NH<sub>3</sub> remaining in this solution.

Since ammonia is a weak base, **the ionisation of ammonia requires energy** as some of the energy is consumed in breaking the N-H bonds.

HCl is a **strong acid** (B-L proton donor) and as per B-L theory, HCl **donates its proton** and becomes Cl<sup>-</sup>.

Hence, when HCl is added to this solution, **most of the HCl will react directly with the NH<sub>3</sub>, instead of OH<sup>-</sup>**.

NH<sub>3</sub> **accepts the proton** and becomes NH<sub>4</sub><sup>+</sup>. Hence, the reaction equation will be:



This leads to a **different chemical reaction** than the one referenced in the enthalpy of neutralisation.

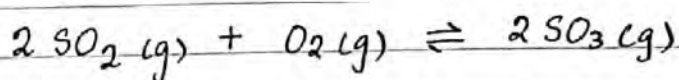
Hence, the value of enthalpy of this reaction will be the sum of  $\Delta H_1$  and  $\Delta H_2$  and therefore **different**.

**Note:** There is no requirement to state if the value is higher or lower since it can only be determined by complex thermodynamic calculation (beyond the HSC syllabus).

Question 5

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Shows correct relationship for <math>Q = K_{eq} = 1</math> when valve is closed</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |
| <ul style="list-style-type: none"> <li>Uses stoichiometric relationship for the chemical equation to derive <math>Q = K_{eq} = 1</math> and/or writes a balanced chemical equation<br/>AND</li> <li>Identifies TWO changes when valve is opened</li> <li>AND uses <u>correct analysis</u> (principles of validity and lack of sufficient information) to and reject LCP and collision theory<br/>AND</li> <li>Choose the most appropriate principle as comparison of <math>Q</math> and <math>K</math> <u>to derive</u><br/><math>Q_{before} = Q_{after} = K_{eq}</math><br/>AND</li> <li>Correct conclusion: Position of <math>K_{eq}</math> does not change</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Uses stoichiometric relationship for the chemical equation to derive <math>Q = K_{eq} = 1</math> and/or writes a balanced chemical equation<br/>AND/OR</li> <li>Identifies TWO changes when valve is opened<br/>AND uses <u>correct analysis</u> (principles of validity and lack of sufficient information) to and reject LCP and Collision Theory<br/>AND</li> <li>Choose the most appropriate principle as comparison of <math>Q</math> and <math>K</math> <u>to derive</u><br/><math>Q_{before} = Q_{after} = K_{eq}</math><br/>AND</li> <li>Correct conclusion: Position of <math>K_{eq}</math> does not change</li> </ul>   | 3     |
| <ul style="list-style-type: none"> <li>Uses stoichiometric relationship for the chemical equation to derive <math>Q = K_{eq} = 1</math> and/or writes a balanced chemical equation<br/>AND/OR</li> <li>Identifies TWO changes when valve is opened AND uses <u>correct analysis</u> (principles of validity and lack of sufficient information) to and reject LCP and collision theory<br/>AND/OR</li> <li>Choose the most appropriate principle as comparison of <math>Q</math> and <math>K</math> <u>to derive</u><br/><math>Q_{before} = Q_{after} = K_{eq}</math><br/>AND</li> <li>Correct conclusion: Position of <math>K_{eq}</math> does not change</li> </ul>    | 2     |
| <ul style="list-style-type: none"> <li>Two partial correct statements<br/>OR</li> <li>One complete correct statement</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |

Sample Response



$n$       2            :      1            :      2            when  $V = 1\text{L}$

Valve closed: Let number of moles be at above ratio

$$Q_{\text{Before}} = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]} = \frac{\left(\frac{2}{1}\right)^2}{\left(\frac{2}{1}\right)^2(1)} = 1$$

$Q_{\text{Before}} = K_{\text{eq}} = 1$  at any point of time

When valve is opened two changes occur.

(1) Volume of vessel  $1\text{L} \rightarrow 3\text{L}$  so concentration of all gases decrease to  $\frac{1}{3}$ .

(2) Extra 2 moles of  $\text{O}_2$  is introduced, ~~so~~ the concentration ~~becomes~~  $[\text{O}_2] = \frac{3 \text{ moles}}{3\text{L}} = 1\text{M}$  (unchanged)

[Note: Addition of reactant must occur at constant volume]

**Validity** • Hence LCP cannot be applied because this principle works for one imposed change only at a time.

OR

• Relative rates of reaction cannot be applied, due to insufficient information. Increase in volume decreases all conc of gases, but the extent of reduction for each reaction cannot be determined.

• Comparing  $Q$  and  $K$  as follows: (Best to use)

Tap opened:  $[\text{SO}_2]_{\text{After}} = \frac{2}{3}$      $[\text{SO}_3] = \frac{2}{3}$      $[\text{O}_2] = \frac{3}{3} = 1$

$$Q_{\text{After}} = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2[\text{O}_2]} = \frac{\left(\frac{2}{3}\right)^2}{\left(\frac{2}{3}\right)^2 \cdot 1} = 1$$

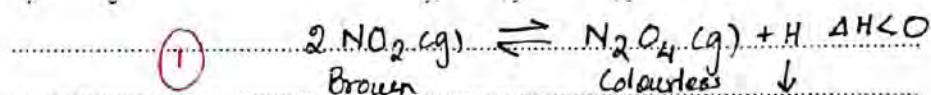
So  $Q_{\text{Before}} = Q_{\text{After}} = K_{\text{eq}}$

(i) Position of equilibrium does not change.

# Question 6

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Uses given stimuli (table and image) to <u>explain</u>.</li> <li><u>Reversibility</u>: using K value to derive forward reaction favoured<br/> <math>K_{eq} = 3.8 \times 10^{29}</math> (large) <math>N_2O_4</math> favoured<br/> AND</li> <li><u>Enthalpy</u>: uses the equation <math>2NO_2(g) \rightleftharpoons N_2O_4(g)</math> and the given information to derive that forward reaction is exothermic <math>\Delta H &lt; 0</math><br/> AND</li> <li>Links entropy to spontaneity: Lower entropy, lower disorder, lower temperature (<math>-196^\circ C</math>) <math>\Delta G &lt; 0</math>, reaction spontaneous<br/> AND</li> <li>Maintains chronological order</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Any TWO explained well and no reference to stimuli.<br/> OR</li> <li>All three outlined with detail missing.<br/> OR</li> <li>All three explained well but no reference to spontaneity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 2     |
| <ul style="list-style-type: none"> <li>Anyone explained well as above<br/> OR</li> <li>Errors in conceptual understanding in one or more aspects</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1     |

## Sample Response



when  $T = 100$ ,  $\Delta G > 0$ ,  $K_{eq}$  is very small. This is a non spontaneous reaction that favours reactant ( $NO_2$ ) and requires external heat energy to drive the reaction forward.

System will be at equilibrium rate of formation of  $NO_2$  is equal to rate of formation of  $N_2O_4$  when  $K_{eq} = 1$

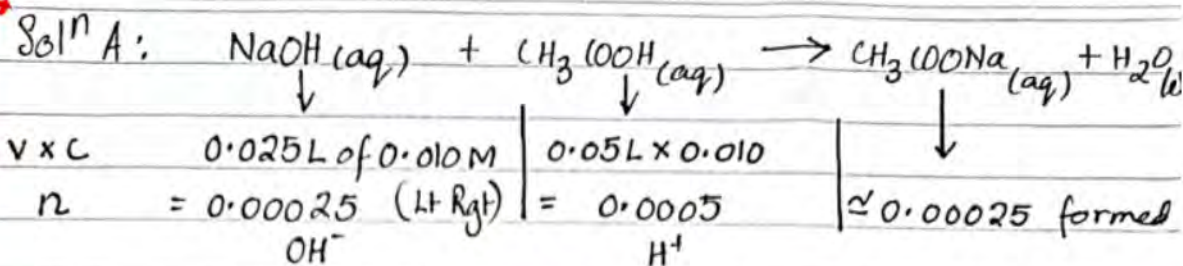
As  $T$  is lowered to  $-196^\circ C$ , reaction becomes spontaneous

① and  $K_{eq}$  increases to  $3.8 \times 10^{29}$  indicating shift of equilibrium to favour products  $N_2O_4$  (colourless). Increase in  $[N_2O_4]$  proves forward reaction is exothermic ( $\Delta H < 0$ ) as system tries to heat up the system (LeP) by favouring the exothermic side.

① Entropy: lowering  $T = -196$  reduces entropy as shift occurs from 2 gaseous moles to 1 gaseous mole leading to  $\Delta G < 0$  a spontaneous reaction

## Question 7

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Shows <b>deep understanding of buffers</b></li> <li>AND</li> <li>Uses a chemical equation and relevant calculations to <b>explain in detail</b> solution A is as buffer</li> <li>AND</li> <li><b>Uses BL theory and LCP</b> and links relevant chemical equations to <b>explain</b> the role of buffer (<math>\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}</math>) when a small amount of acid (<math>\text{H}_3\text{O}^+</math>) is added and its resistance to change in pH when <math>\text{pH} = -\log[\text{H}^+]</math> AND</li> <li>Justifies with relevant calculations, relevant equation with <b>reference to BL theory and/or</b> LCP why solution B is NOT a buffered solution in detail</li> <li>AND</li> <li>Explains the <b>hydrolysis of basic salt</b> <math>\text{CH}_3\text{COONa}</math> for solution B and its drastic drop in pH when small amount of acid is added AND compares the change in pH when small amount of acid is added.</li> </ul>                                               | 5     |
| <ul style="list-style-type: none"> <li>Student shows <b>very good understanding of buffers</b></li> <li>AND</li> <li>Uses a chemical equation and relevant calculations <b>to explain</b> solution A is as buffer</li> <li>AND</li> <li>Uses BL theory <b>OR</b> LCP and links relevant chemical equations to explain the role of buffer (<math>\text{CH}_3\text{COOH}/\text{CH}_3\text{COONa}</math>) when a small amount of acid (<math>\text{H}_3\text{O}^+</math>) is added and its resistance to change in pH when <math>\text{pH} = -\log[\text{H}^+]</math></li> <li>AND</li> <li>Justifies with relevant calculations, relevant equation with reference to BL theory and/or LCP why solution B is NOT a buffered solution</li> <li>AND</li> <li><b>Provides an outline</b> the hydrolysis of basic salt <math>\text{CH}_3\text{COONa}</math> for solution B and its drastic drop in pH when small amount of acid is added.</li> <li><b>(Detail lacking, 1 chemical equation only, no calculation and hydrolysis not mentioned no error)</b></li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Detail lacking for explanation for solution A OR solution B AND/OR one error</li> <li>AND</li> <li>Limited chemical equations but correct calculations</li> <li>AND/OR</li> <li>Limited OR No calculation</li> <li>AND</li> <li>Hydrolysis not addressed</li> <li>OR</li> <li>Solution explained in detail but solution B outlined only or vice versa</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 3     |
| <ul style="list-style-type: none"> <li>Detail Lacking and limited equations OR calculations OR</li> <li>No equations and/or no calculation OR</li> <li>No explanation in reference to LCP or BL theory AND pH not determined using <math>\text{pH} = -\log[\text{H}^+]</math></li> <li>AND/OR Hydrolysis not addressed</li> <li>OR Incorrect explanation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2     |
| <ul style="list-style-type: none"> <li>Any one correct step (either one calculation, one explanation OR one chemical equation)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 1     |



excess  $\text{CH}_3\text{COOH}$  left in sol<sup>n</sup> = 0.00025

Note: % ionisation has been ignored as value not provided.

0.00025 moles NaOH will neutralise  $\text{CH}_3\text{COOH}$  and produce 0.00025 moles of  $\text{CH}_3\text{COONa}$  in given solution.  
Total volume = 0.075 L

$$[\text{CH}_3\text{COONa}] = \frac{0.00025 \text{ moles}}{0.075 \text{ L}} = 0.003 \text{ M}$$

$$[\text{CH}_3\text{COOH}] \approx \frac{0.00025 \text{ (excess)}}{0.075 \text{ L}} = 0.003 \text{ M}$$

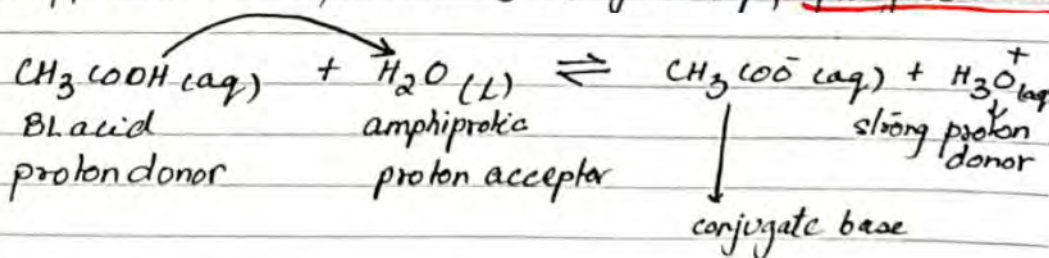
①

Since weak acid  $\text{CH}_3\text{COOH}$  and its conjugate base  $\text{CH}_3\text{COO}^-$  (salt) are in equimolar ratio, Sol<sup>n</sup> A is a buffer solution. (0.003M)

When a small amount of acid ( $\text{H}^+$ ) is added to the buffer solution, the  $[\text{H}_3\text{O}^+]$  goes up, pH goes down.

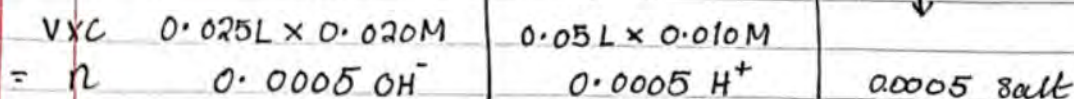
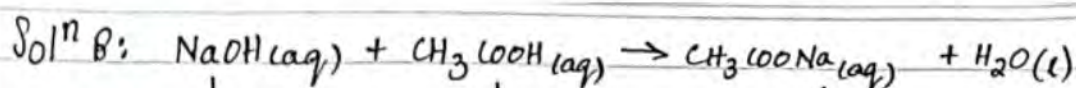
①

BLTh



Acc to LCP, equilibrium shifts left & favours the formation of  $\text{CH}_3\text{COOH}$  (reverse reaction) in order to reduce  $[\text{H}_3\text{O}^+]$  & counter the imposed change. Thus it resists further decrease in pH & the pH is maintained in a small range.

①



0.0005 moles of CH<sub>3</sub>COOH will react with 0.0005 moles of NaOH to produce 0.0005 moles of the salt CH<sub>3</sub>COONa with no excess CH<sub>3</sub>COOH left in sol<sup>n</sup>.

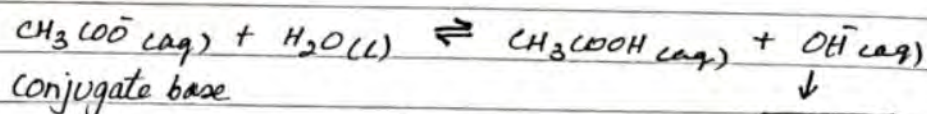
So after the neutralisation is complete solution B is a basic solution with a pH > 7 due to the basic salt CH<sub>3</sub>COONa.

$$[\text{CH}_3\text{COONa}] = \frac{0.0005 \text{ moles}}{0.075 \text{ L}} = 0.006 \text{ M}$$

① Since there is no corresponding weak acid left this is not a buffer solution. (Sol B)

CH<sub>3</sub>COONa dissociates in water as:  $\text{CH}_3\text{COONa(aq)} \rightarrow \text{CH}_3\text{COO}^-(\text{aq}) + \text{Na}^+(\text{aq})$

Hydrolysis of CH<sub>3</sub>COO<sup>-</sup> in water can be written as

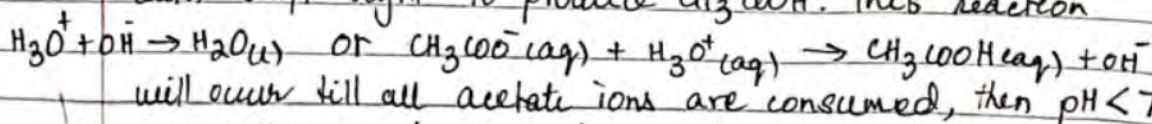


Here  $K_b > K_a$

pH > 7

hence a basic solution

① when a small amount of acid (H<sub>3</sub>O<sup>+</sup>) is added (say x) it will react with x amount of CH<sub>3</sub>COO<sup>-</sup> & equilibrium will shift right to produce CH<sub>3</sub>COOH. This reaction



will occur till all acetate ions are consumed, then pH < 7

→ as small amounts of acid are added pH will drop drastically.

## Question 8(a)

| Marking Criteria                                                                                                                                                                                                                                                                                                  | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>All relevant and correct calculations shown in detail<br/>AND</li> <li>One chemical equation relevant to back titration mole ratio<br/>AND</li> <li>Claim verified correctly: Invalid claim 16.1% (v/v) instead of 30-32% (v/v).</li> </ul>                                | 7     |
| <ul style="list-style-type: none"> <li>All relevant and correct calculations shown in detail BUT ONE error<br/>AND</li> <li>One chemical equation relevant to back titration mole ratio<br/>AND</li> <li>Claim verified correctly: Claim justified appropriately and supported by above calculations.</li> </ul>  | 6     |
| <ul style="list-style-type: none"> <li>All relevant and correct calculations shown in detail BUT TWO errors<br/>AND</li> <li>One chemical equation relevant to back titration mole ratio<br/>AND</li> <li>Claim verified correctly: Claim justified appropriately and supported by above calculations.</li> </ul> | 5     |
| <ul style="list-style-type: none"> <li>More than two errors and relevant calculations NOT shown in detail<br/>AND</li> <li>Claim justified appropriately and supported by above calculations.<br/>OR</li> <li>All relevant steps &amp; equations NOT shown.</li> </ul>                                            | 4     |
| <ul style="list-style-type: none"> <li>Multiple errors in calculations and relevant steps missed AND equation not provided<br/>AND</li> <li>Invalid claim.</li> </ul>                                                                                                                                             | 3     |
| <ul style="list-style-type: none"> <li>Any two correct steps with one chemical equation AND/OR valid claim based on calculations.</li> </ul>                                                                                                                                                                      | 2     |
| <ul style="list-style-type: none"> <li>Any one relevant calculation correctly shown such as average titre = 0.02962L<br/>OR</li> <li>One correct and relevant chemical equation.</li> </ul>                                                                                                                       | 1     |

| Sample | Volume of sodium thiosulfate (mL) |   |
|--------|-----------------------------------|---|
| A      | 31.92                             | X |
| B      | 29.63                             | ✓ |
| C      | 29.61                             | ✓ |
| D      | 29.62                             | ✓ |

- (a) Given the density of ethanol is  $0.789 \text{ g mL}^{-1}$ , does the sample verified meet the manufacturer's claim?

7

Support your answer at least one relevant equation and all relevant calculations.

$$1. \text{ titre ave} = (29.63 + 29.61 + 29.62) \div 3 = 29.62 \text{ mL} = \boxed{0.02962 \text{ L}} \quad (1)$$

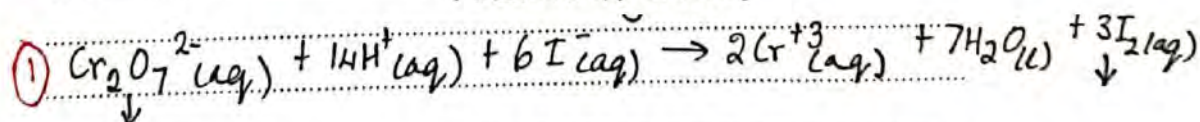
$$2. n_{\text{Na}_2\text{S}_2\text{O}_3} = C \times V = 0.010 \frac{\text{mol}}{\text{L}} \times 0.02962 \text{ L} = 0.0002962$$

$$= 2.962 \times 10^{-4} \text{ moles}$$

$$n_{\text{S}_2\text{O}_3^{2-}} = 2.962 \times 10^{-4} \text{ moles}$$

$$\text{So } n_{\text{I}_2} = \frac{1}{2} \times n_{\text{S}_2\text{O}_3^{2-}} = 0.0001481 \text{ moles}$$

$$\text{During titration } (1) n_{\text{Cr}_2\text{O}_7^{2-}} (\text{remaining or excess}) = \frac{1}{3} \times 0.0001481 = \boxed{4.94 \times 10^{-5}}$$



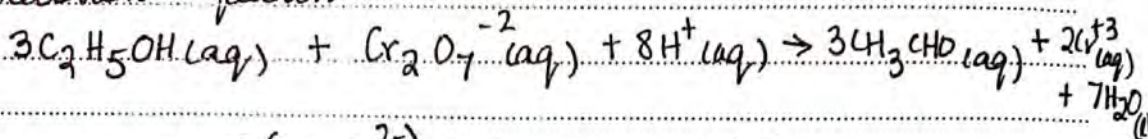
$$\text{Initial Moles of } \text{Cr}_2\text{O}_7^{2-} \text{ added to diluted wine at the start} = C \times V$$

$$= 0.040 \frac{\text{mol}}{\text{L}} \times 0.02 \text{ L} = 0.0008 \text{ moles}$$

$$(1) \text{ So moles of } \text{Cr}_2\text{O}_7^{2-} \text{ reacted with EtOH} = \text{Initial} - \text{Remaining}$$

$$= 0.0008 - 0.0000494 = \boxed{7.51 \times 10^{-4}}$$

Relevant Equation :

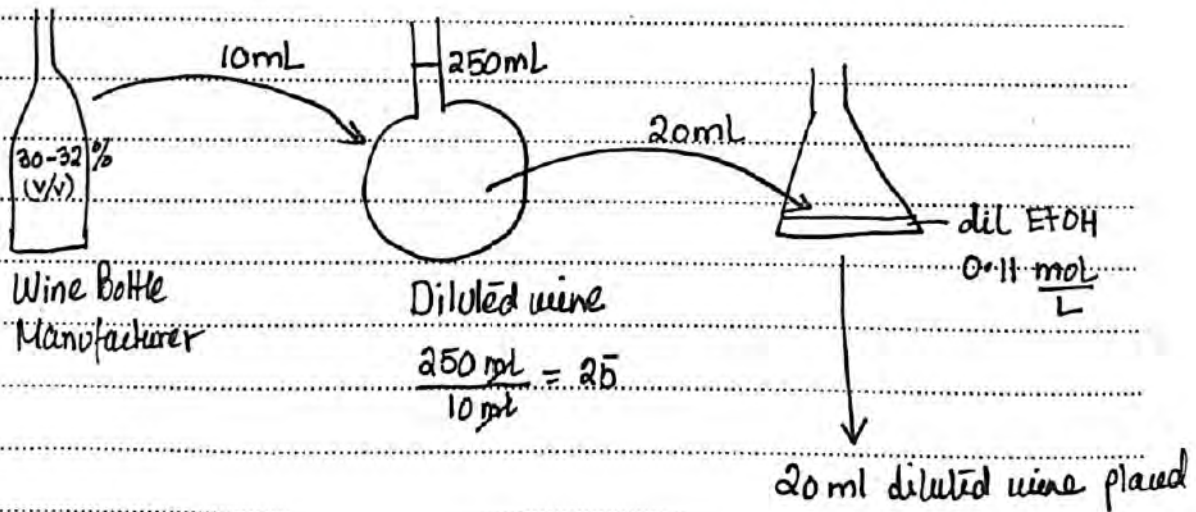


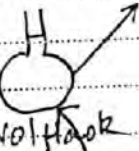
$$n(\text{Cr}_2\text{O}_7^{2-}) : n_{\text{EtOH}} = 1 : 3$$

$$n_{\text{EtOH}} = 3 \times 7.51 \times 10^{-4} \text{ moles}$$

$$= \boxed{2.25 \times 10^{-3} \text{ moles C}_2\text{H}_5\text{OH}} \quad (1)$$

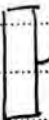
## Working Out for dilution



 20 ml EtOH contains  $2.25 \times 10^{-3}$  moles  
 250 ml Vol. flask "  $2.25 \times 10^{-3} \times \left(\frac{250}{20} = 12.5\right)$

Vol. flask

$$250 \text{ ml} = 0.028125 \text{ moles}$$

 10ml  
 Wine bottle

10ml EtOH was transferred to vol. flask then topped up to 250 ml with  $\text{dH}_2\text{O}$

Implies # of moles from wine bottle (10ml) to vol. flask (250ml) is the same.

$\Rightarrow$  10 ml contains 0.028125 moles (1)

$$M_w(C_2H_5OH) = 12.01 \times 2 + 1.008 \times 6 + 16 = 46.07 \frac{g}{mol}$$

$$n = \frac{mass}{M_w} \Rightarrow n \times M_w = mass$$

$$= 0.028125 \times 46.07 = 1.296g$$

$$Mass = Vol \times density \Rightarrow density = \frac{mass}{Volume}$$

$$Volume = \frac{1.296g}{0.789 g ml^{-1}}$$

So 10 mL (bottle) contains 1.64 mL EtOH = 1.64 mL

Then 100 mL (bottle) "  $\frac{1.64}{10} \times 100 = 16.4\% (v/v)$  (1)

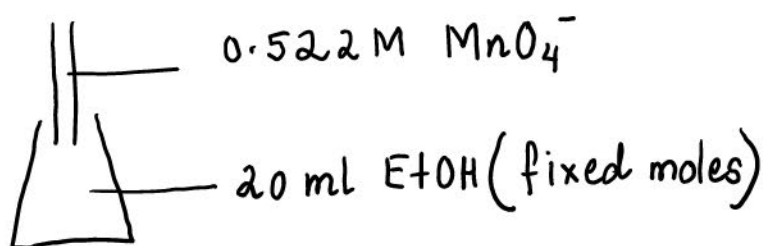
Question 8(b)

| Marking Criteria                                                                                                                                                                                                                                                          | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Two observations (O1, O2) linked to inaccuracies in data and provides detailed improvement with justification for both errors in methodology (Im)</li> <li>Succinct well explained (Ex), maintains chronological order.</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Any one of the two observations AND explains inaccurate data in detail and outlines one improvement.</li> </ul>                                                                                                                    | 3     |
| <ul style="list-style-type: none"> <li>One relevant observation outlined with some justification and improvement</li> </ul>                                                                                                                                               | 1–2   |

Sample Response

**Observation 1**

Bubbles were visible during the titrations, which may be attributed to the formation of  $\text{CO}_2$  in the conical flask from the oxidation of ethanoic acid to carbon dioxide, thus inhibiting the reaction of  $\text{KMnO}_4$  with ethanol. For the complete neutralisation of ethanol to occur, greater amount of permanganate, and therefore a greater titre volume, would be required to react with all the ethanol moles in the conical flask and reach the endpoint (slightly higher than equivalence point).



**Inaccurate data 1:** Higher titre volume will lead to a higher calculated value of EtOH content thus overestimating the manufacturer's claim.

**Observation 2**

The wine may contain  $\text{Mn}^{2+}$  which acts as a **catalyst**, thus can cause an **incomplete reduction of  $\text{MnO}_4^-$**  ions. A brown solid  $\text{MnO}_2$  was formed in titration #3. This implies the  $\text{Mn}^{2+}$  catalyst speeds up the reaction and reacts faster with the permanganate ions. Since these  $\text{MnO}_4^-$  ions have been incompletely reduced, a **smaller amount of ethanol** would have been oxidised by these  $\text{MnO}_4^-$  ions.

**Inaccurate data 2:** This means a greater amount of  $\text{MnO}_4^-$  ions would have been required to react with all the ethanol, which would explain the very high titre volume obtained.

**Observation 3**

The reaction rate increased after addition of the first drop of titrant. The endpoint itself occurs when there is excess permanganate in the conical flask, it would occur just after the equivalence point.

**Inaccurate data 3:** Using **one full drop** for each titration rather than **use of half-drops** in the methodology towards the end of the titration, **increases the risk of overshooting** the endpoint.

**Improvements to the methodology:**

1. If  $\text{CO}_2$  formation had occurred, this issue could be rectified by the use of a weaker oxidising agent than permanganate ions, one that would not react with ethanoic acid (but would with ethanol) such as  $\text{K}_2\text{Cr}_2\text{O}_7$ . This would remove the systematic error caused by the presence of very strong oxidising agent.
2. By changing the oxidising agent, the catalyst effect and formation of brown solid can be avoided.
3. The use of half-drops (or even quarter-drops) would decrease the risk of overshooting the endpoint.

# Normanhurst Boys High School

## 2023 Year 12 Chemistry Trial Examination

### Marking Responses, Sample Answers & Notes from the Marker



#### Section 2C (20 marks)

#### Written Response

##### Question 1(a)

| Marking Criteria                                                                                                                                                                                                                             | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"><li>Determines the correct molecular formula WITH appropriate working and reasoning.</li></ul>                                                                                                             | 2     |
| <ul style="list-style-type: none"><li>Determines the correct molecular formula WITHOUT appropriate working and reasoning.</li></ul> OR <ul style="list-style-type: none"><li>States the correct molecular formula without working.</li></ul> | 1     |

##### Sample Response

The empirical formula of the compound is:

|             | C                                      | H                                        | O                             |
|-------------|----------------------------------------|------------------------------------------|-------------------------------|
| Mass (g)    | 69.7                                   | 11.7                                     | 18.6                          |
| Moles (mol) | $\frac{69.7}{12.01} = 5.8035\dots$     | $\frac{11.7}{1.008} = 11.6071\dots$      | $\frac{18.6}{16.00} = 1.1625$ |
| Mole ratio  | $\frac{5.8035\dots}{1.1625} \approx 5$ | $\frac{11.6071\dots}{1.1625} \approx 10$ | $\frac{1.1625}{1.1625} = 1$   |

Since the empirical formula has a molecular mass of 86, and the molecular ion peak has a charge to mass ratio of 86, the molecular formula of the compound is  $C_5H_{10}O$ .

##### Notes from the Marker

- Better responses were able to determine the empirical formula and demonstrated evidence of correctly using the mass spectrum data to determine the molecular formula of the unknown compound.
- Weaker responses assumed that the  $m/z$  of the  $M+1$  peak was the molar mass of the compound. It should be noted that the  $M+1$  peak is caused by the presence of the  $^{13}C$  isotope in the molecule.

Question 1(b)

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Determines plausible TWO possible isomers and excludes ONE possible isomer with ALL of the following features, with extensive detail:               <ul style="list-style-type: none"> <li>interprets the mass spectrum data and proposes how this relates to the isomers</li> <li>interprets the IR spectrum data and proposes how this correlates to the isomers AND/OR excludes other isomers</li> <li>interprets chemical tests and proposes how this correlates to the isomers AND excludes other isomers</li> <li>proposes TWO possible isomers and excludes ONE isomer based on ALL of the valid interpretation of the proposed data</li> </ul> </li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>Determines TWO possible isomers and excludes ONE possible isomer but makes ONE error. OR</li> <li>Determines plausible TWO possible isomers and excludes ONE possible isomer with ALL of the above features with a good level of detail.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                | 3     |
| <ul style="list-style-type: none"> <li>Determines ONE possible isomer and excludes ONE possible isomer but makes ONE error with appropriate reasoning. OR</li> <li>Determines plausible TWO possible isomers and excludes ONE possible isomer with TWO of the above features with some reasoning.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                              | 2     |
| <ul style="list-style-type: none"> <li>Provides some relevant information.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 1     |

Sample Response

The three possible structures for the compound are pentanal, pentan-2-one and pentan-3-one. This is because the:

- sharp IR peak at  $1720\text{ cm}^{-1}$  corresponds to a C=O bond which suggests that the compound could be an aldehyde or a ketone
- molecular ion peak has a  $m/z$  of 86 which corresponds to a molar mass of approximately  $86\text{ g mol}^{-1}$
- fragment ion peak at  $m/z = 28$  has come from a fragment such as  $[\text{C}=\text{O}]^+$

However, based on the chemical tests using Fehling's reagent and Tollen's reagent, both of the negative results suggest that the unknown compound is a ketone, as an aldehyde would lead to a positive result due to its oxidation to pentanoic acid. Hence, the unknown compound could be pentan-2-one and pentan-3-one and not pentanal.

Notes from the Marker

- Better responses were able to utilise all of the data in order to deduce potential isomers and to rule out one possible isomer. Students are reminded
- Weaker responses assumed that the  $m/z$  of the  $M+1$  peak was the molar mass of the compound. It should be noted that the  $M+1$  peak is caused by the presence of the  $^{13}\text{C}$  isotope in the molecule.

## Question 2

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Extensively explains THREE relevant factors with reference to how that factor benefits the production of nitric acid.</li> <li>Makes specific reference to the flow chart for EACH factor.</li> </ul>                                                                                                                                                                               | 4     |
| <ul style="list-style-type: none"> <li>Explains TWO relevant factors with some reference to the flow chart.</li> </ul> OR <ul style="list-style-type: none"> <li>Explains ONE relevant factor and outlines TWO other relevant factors with some reference to the flow chart.</li> </ul> OR <ul style="list-style-type: none"> <li>Explains THREE relevant factors without specific reference to the flow chart.</li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Explains ONE relevant factor.</li> </ul> OR <ul style="list-style-type: none"> <li>Outlines TWO relevant factors.</li> </ul>                                                                                                                                                                                                                                                        | 2     |
| <ul style="list-style-type: none"> <li>Provides some relevant information.</li> </ul>                                                                                                                                                                                                                                                                                                                                      | 1     |

### Sample Response

#### Use of a catalyst

The use of a Pt catalyst in the catalytic chamber allows the reaction to proceed at lower temperatures as it provides an alternate pathway with a lower activation energy. Combined with the high temperature and high pressure, the rate of reaction is increased to convert the ammonia and oxygen gas into nitric oxide and water vapour due to the increased likelihood of successful collisions between reactant molecules. This maximises the efficiency of this step in the production of nitric acid as the bulk of the reactants have been converted into product, thus reducing the running costs of this chamber as it is in operation for a shorter period of time.

#### Use of a heat exchanger

The use of a heat exchanger enables the reaction mixture to be cooled down as the reaction in the catalytic chamber is very exothermic and poses a safety risk if it is not managed. In addition to this, the oxidation chamber utilises low temperatures. By Le Chatelier's principle, when the closed system is disturbed through the use of low temperature (50°C), this favours the forward reaction and the production of nitrogen dioxide in order to counteract this change, thus maximising the yield for this step in the process.

#### Recycling

Unused reactant gases are recycled after being separated from the reaction mixture. This means that resources are not wasted – making the process more economical as less reactant needs to be produced in the catalytic chamber and the oxidation chamber. This also ensures that these gases are not released into the environment as they have the potential to cause pollution (e.g. contribute to acid rain and photochemical smog).

#### Access to reactants

The plant would be located so that ammonia and oxygen gases would be easily accessible for use in the catalytic chamber – perhaps near a Haber process plant or road/rail network so that these resources could be transported easily and cheaply. This ensures that profits can be maximised to ensure that the production of nitric acid is economically viable.

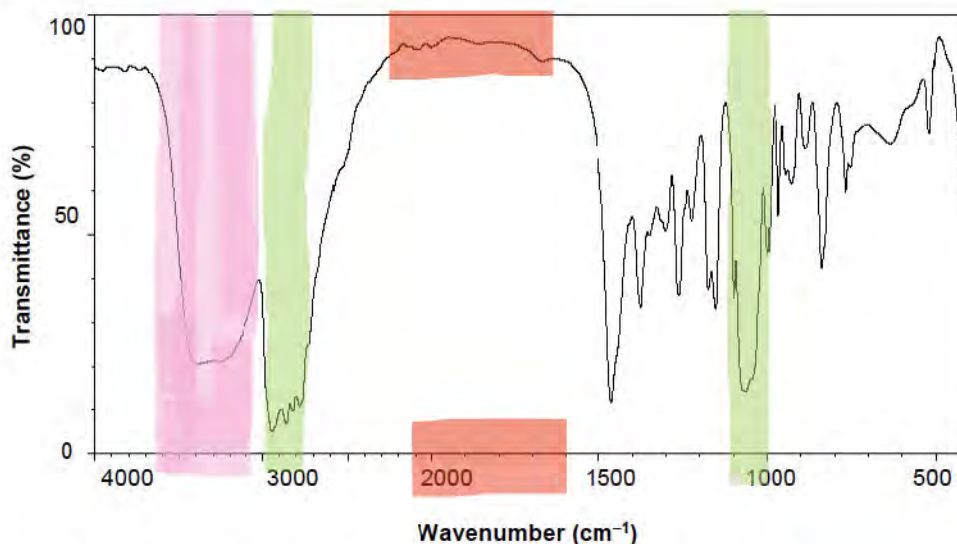
#### Notes from the Marker

- Better responses were able to make explicit reference to the flowchart and link the relevant chemical principles, where appropriate, to the viability of the production of nitric acid in an industrial context.
- Weaker responses assumed links to chemical principles was sufficient and did not explain how their proposed factors were necessary considerations to feasibly produce nitric acid.

Question 3

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Writes the correct structural formula for the given compound (<b>naming not required</b>).</li> <li>Justifies the correct structure showing an extensive understanding of the interpretation of spectroscopic data and refers explicitly to the relevant spectroscopic data.</li> <li>Infrared spectrum               <ul style="list-style-type: none"> <li>TWO peaks shown are addressed in terms of bonds.</li> <li>Lack of a peak due to the C=O bond addressed to rule out carboxylic acid.</li> </ul> </li> <li><sup>13</sup>C NMR spectrum               <ul style="list-style-type: none"> <li>Number of carbon environments addressed in relation to the molecular formula and symmetry.</li> <li>All peaks addressed.</li> </ul> </li> <li><sup>1</sup>H NMR spectrum               <ul style="list-style-type: none"> <li>Number of hydrogen environments addressed in relation to the molecular formula.</li> <li>All peaks addressed in relation to data, (n+1) rule and relative peak area with sufficient detail.</li> </ul> </li> </ul> | 7     |
| <ul style="list-style-type: none"> <li>Writes the correct structural formula for <b>the given compound (naming not required)</b> or a structure very close to the correct structure.</li> <li>Justifies the structure showing a thorough understanding of the interpretation of spectroscopic data.</li> <li>Refers to relevant spectroscopic data.</li> </ul> <p>OR</p> <ul style="list-style-type: none"> <li>Deduces the correct molecule and provides thorough analysis of the spectroscopic data BUT <b>some critical peak analysis missing</b>.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 6     |
| <ul style="list-style-type: none"> <li>Shows a sound understanding of the interpretation of spectroscopic data with <b>critical analysis of data missing</b>.</li> <li>Uses relevant information presented in the question to justify the structure of the chemical.</li> <li>Provides a structural formula consistent with analysis.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4–5   |
| <ul style="list-style-type: none"> <li>Shows some understanding of the interpretation of spectroscopic data.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 2–3   |
| <ul style="list-style-type: none"> <li>Provides some relevant information.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |

## Sample Response

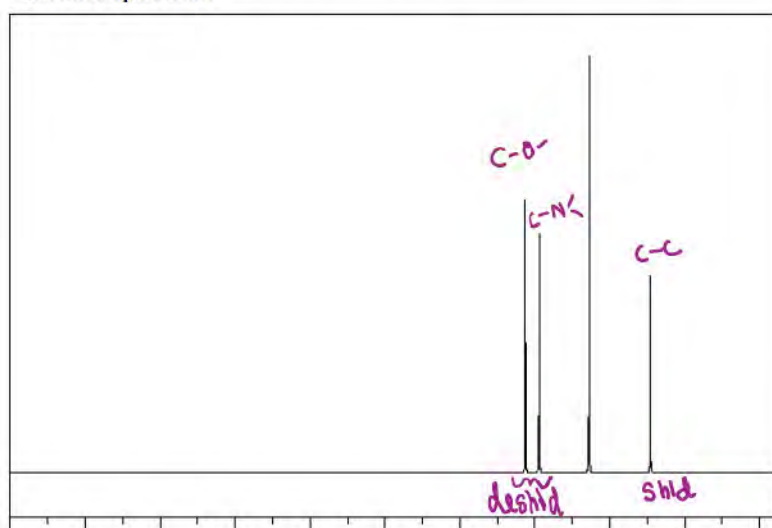


From the infrared spectrum, the:

- **strong and broad peak between 3230 - 3500 cm<sup>-1</sup>** suggests that there is a **O-H bond** and /or N-H bond in the structure.
- **Absence of a strong and broad peak at 1733 cm<sup>-1</sup>** suggests that the organic molecule **does not** contain a **C=O bond** in the structure, hence the molecule may not contain any carbonyl group due to a carboxylic acid functional group or ketones.
- **Fingerprint region (less than 1500 cm<sup>-1</sup>)** cannot be conclusively analysed due to the overlapping of peaks due to different bonds, however there is a clear and strong absorption at 1000 cm<sup>-1</sup>. This could be possibly due to a **C-O bond**.

2

## <sup>13</sup>C NMR spectrum

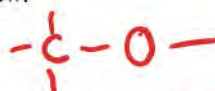


From the <sup>13</sup>C spectrum, there are four unique carbon environments. As there are 5 C atoms, this suggests that two C atoms are in the same environment due to symmetry in a mirror plane.



The peak(s) at:

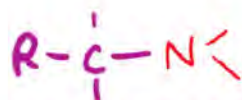
- The **peak at 63 ppm** to the extreme left has the **highest chemical shift** and corresponds to a C atom single bonded to an O atom as shown below.



This is **most de-shielded** due to the carbon next to **highly electronegative O atom** resulting in the peak being downfield.

- Peak at 60 ppm corresponds to a C atom singly bonded to an N atom as shown below.

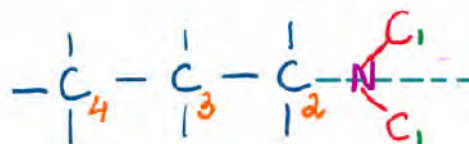
1



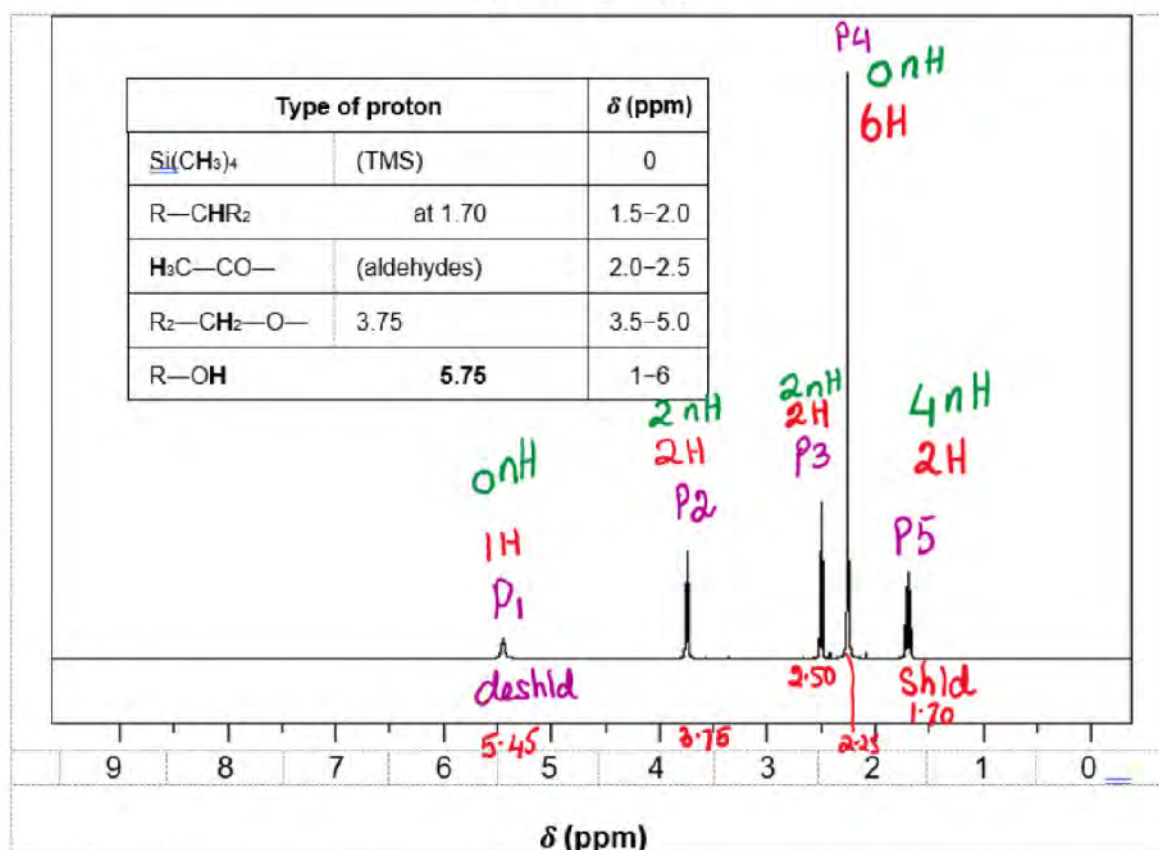
This is de-shielded as well due to the **carbon next to the electronegative N atom** resulting in the peak being reasonably downfield. (not required use this term in Chemistry exam)

- Peak at 46 ppm corresponds to **C atom bonded to the N in an alkyl group** which is consistent with chemical shift between 25–60 ppm.
- Peak at 29 ppm correspond to **C atoms bonded to the C in an alkyl group** which is consistent with chemical shift between 5–40 ppm.

Possible structure so far:



### <sup>1</sup>H NMR spectrum



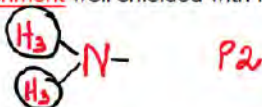
From the  $^1\text{H}$  spectrum, there are **five unique hydrogen environments**. By considering the molecular formula, the **number of H atoms in each environment corresponds to the relative peak area in a 1:1 ratio**.

Through the use of the  $n+1$  rule to analyse the splitting patterns and based on the relative peak area, the:

- **Singlet at 5.45 ppm** indicates that there are **no neighbouring H atoms**, relative peak area of 1 corresponds to **one H atom** in the following environment. It is highly de-shielded due to the neighbouring O atom.

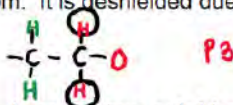


- **Singlet at 2.25 ppm** indicates that there are **no neighbouring H atoms**, the relative peak area of 6 corresponds to **6 H atoms in the environment** well shielded with low chemical shift.

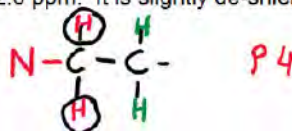


(3)

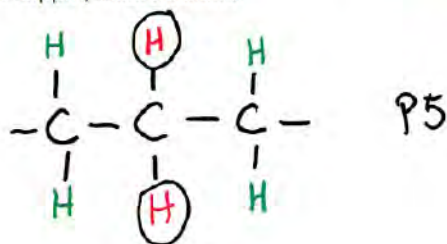
- **triplet peak at 3.75 ppm** indicates that there are **two neighbouring H atoms**, most likely in a neighbouring  $-\text{CH}_2$  group. The relative peak area of 2 corresponds to **2 H atoms** in the environment as the shift occurs between 2.1–2.6 ppm. It is deshielded due to the nearby O atom.



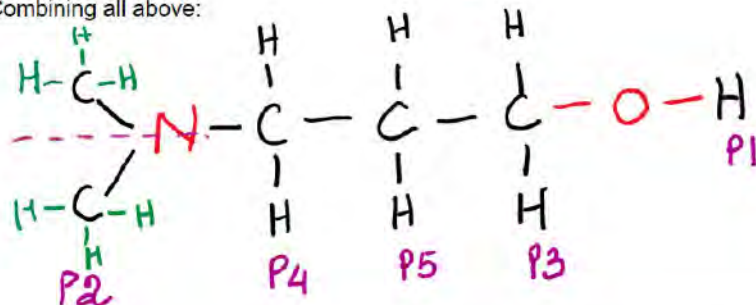
- **triplet peak at 2.50 ppm** indicates that there are **two neighbouring H atoms**, most likely in a neighbouring  $-\text{CH}_2$  group. The relative peak area of 2 corresponds to **2 H atoms** in the environment as the shift occurs between 2.1–2.6 ppm. It is slightly de-shielded due to the nearby N atom.



- **Quintet peak at 1.70 ppm** indicates that there are **four neighbouring H atoms**, most likely in two neighbouring  $-\text{CH}_2$  group. The relative peak area of 2 corresponds to **2 H atoms in the environment** as the shift occurs between 1.5–2.0 ppm, most shielded.



Combining all above:



(1)  
Correct Structure

#### Notes from the Marker

- Better responses had used and explicitly referred to the spectroscopic data to put together the pieces of the puzzle, clearly showed their thinking processes and annotated the spectroscopic data. These responses showed evidence of analysing each peak thoroughly and provided reasoning and explanation for each of the peaks given.
- Weaker responses did not show care when drawing organic molecules and did not show good annotations to the spectra to clearly link this data to H and C environments in the molecule. These responses also did not explain the significance of each peak or explain the relevance of the number of environments clearly. These responses often were not cohesive or relied on partial analysis of the data.

#### Question 4

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Calculates the molar solubility of calcium fluoride in the solution with ALL of following features:               <ul style="list-style-type: none"> <li>determines the concentration of hydroxide ions in solution</li> <li>calculates the calcium ion concentration based on calcium hydroxide</li> <li>calculates the calcium fluoride solubility using the fluoride ion concentration.</li> </ul> </li> </ul> | 3     |
| <ul style="list-style-type: none"> <li>Calculates the molar solubility of calcium fluoride in the solution with TWO of the above features.</li> </ul>                                                                                                                                                                                                                                                                                                    | 2     |
| <ul style="list-style-type: none"> <li>Provides some relevant information by determining the concentration of hydroxide ions in solution.</li> </ul>                                                                                                                                                                                                                                                                                                     | 1     |

#### Sample Response

At pH = 14.0:

$$\text{pH} + \text{pOH} = 14.0$$

$$\text{pOH} = 14.0 - 12.0$$

$$= 2.0$$

$$\therefore [\text{OH}^-] = 10^{-2.0}$$

$$= 1.0 \times 10^{-2} \text{ mol L}^{-1}$$

For  $\text{Ca}(\text{OH})_2$ :

$$K_{sp} = [\text{Ca}^{2+}][\text{OH}^-]^2$$

$$5.02 \times 10^{-6} = [\text{Ca}^{2+}] \times (1.00 \times 10^{-2})^2$$

$$[\text{Ca}^{2+}] = \frac{5.02 \times 10^{-6}}{(1.00 \times 10^{-2})^2}$$

$$= 0.0502 \text{ mol L}^{-1}$$

For  $\text{CaF}_2$ :

$$K_{sp} = [\text{Ca}^{2+}][\text{F}^-]^2$$

$$3.45 \times 10^{-11} = 0.0502 \times [\text{F}^-]^2$$

$$[\text{F}^-] = \sqrt{\frac{3.45 \times 10^{-11}}{0.0502}}$$

$$= 2.621547 \times 10^{-5} \text{ mol L}^{-1}$$

Therefore, the molar solubility of  $\text{CaF}_2$  in this solution is:

$$[\text{CaF}_2] = \frac{1}{2} \times 2.621547 \times 10^{-5}$$

$$= 1.310773 \times 10^{-5}$$

$$= 1.3 \times 10^{-5} \text{ mol L}^{-1}$$

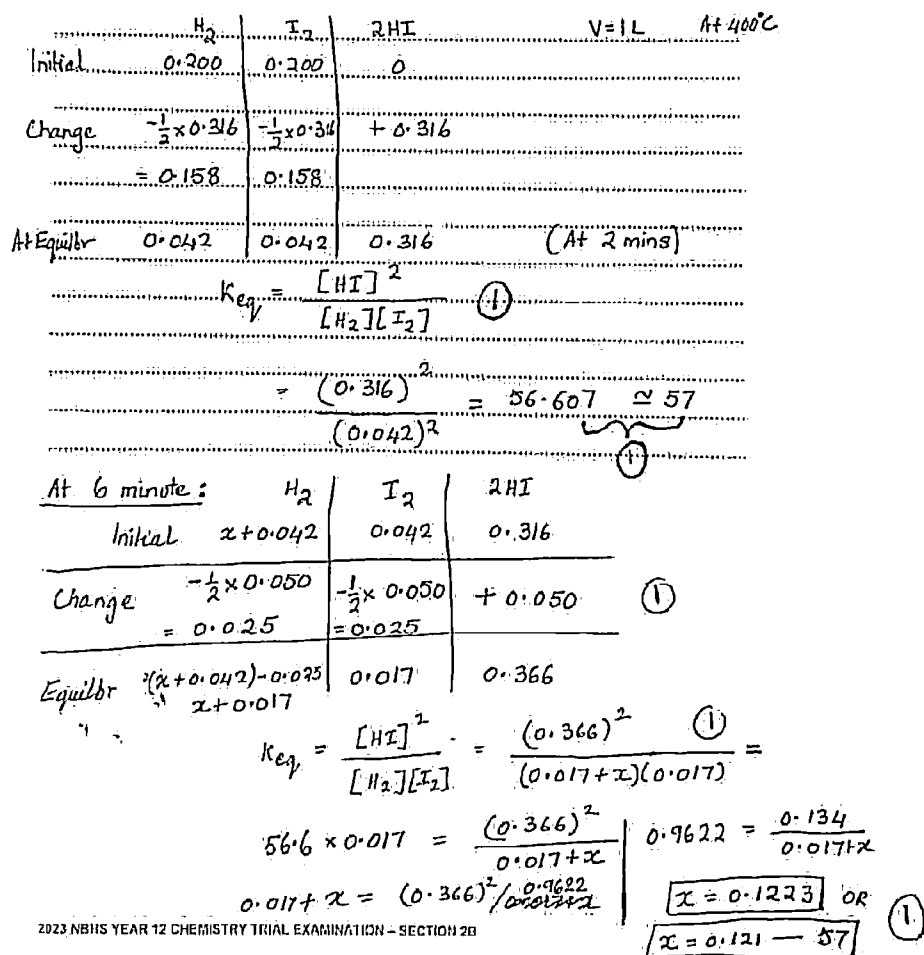
#### Notes from the Marker

- Better responses showed evidence of understanding how the common ion effect can be applied in calculations and ensured that all working out is shown and logically ordered.
- Weaker responses did not utilise the information provided in the question and did not utilise the  $K_{sp}$  value of calcium hydroxide provided on the data sheet.

### Question 3

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math> AND</li> <li>Shows all relevant values for ICE table 2 at 6 minutes AND</li> <li>Sets up correct <math>K_{eq}</math> expression for table 2 AND</li> <li>Solves algebraic equation and calculates the moles of hydrogen added (shown as x). <math>x = 0.1223</math> OR <math>0.121</math>.</li> </ul> | 5     |
| <ul style="list-style-type: none"> <li>All relevant steps correct with any ONE calculation error.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                     | 4     |
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math> AND</li> <li>Error in ICE table 2 AND error in algebraic calculation leading to two significant errors AND</li> <li>Incorrect value for x.</li> </ul>                                                                                                                                                 | 3     |
| <ul style="list-style-type: none"> <li>Uses ICE table 1 accurately AND sets up correct <math>K_{eq}</math> expression AND calculates the value of <math>K_{eq} = 57</math>.</li> </ul>                                                                                                                                                                                                                                                                                           | 2     |
| <ul style="list-style-type: none"> <li>Any one step of calculation or expression correct.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                             | 1     |

### Sample Response



Question 4

| Marking Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>• Uses B-L Theory labels such as proton donor (BL acid) and BL base (proton acceptor) (acid-base chemistry) to explain reaction between strong acids and strong bases<br/>AND</li> <li>• Shows equation for neutralization as <math>\text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l})</math> only for all above neutralisation<br/>AND</li> <li>• Justifies above equation to support <math>\Delta H = -57.6 \text{ kJ mol}^{-1}</math> for all strong acid/base neutralisations<br/>AND</li> <li>• Identifies <math>\text{NH}_3</math> as weak base and explains its ionization in water using B-L theory<br/>AND</li> <li>• Shows ALL relevant equations OR writes all word equations<br/>AND</li> <li>• Explains HCl reacting directly with ammonia instead of <math>\text{OH}^-</math><br/>OR</li> <li>• Explains correctly differently why <math>\text{NH}_3</math> with HCl leads to a different <math>\Delta H</math> value</li> </ul> | 4     |
| <ul style="list-style-type: none"> <li>• Outlines or explains B-L Theory labels to explain reaction between strong acids and strong bases<br/>AND</li> <li>• Outlines/explains the reaction between strong acids such as HCl and <math>\text{NH}_3</math><br/>AND</li> <li>• Uses one/two relevant equations only related to any one of the above</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3     |
| <ul style="list-style-type: none"> <li>• Provides detailed justification for strong acid/base neutralisation with appropriate equation as <math>\text{H}_3\text{O}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow 2\text{H}_2\text{O}(\text{l})</math><br/>AND/OR</li> <li>• Provides correct equation for ammonia and HCl as <math>\text{NH}_3 + \text{HCl} \rightarrow \text{NH}_4^+ + \text{Cl}^-</math> but no explanation for ammonia and HCl is provided<br/>OR</li> <li>• Incorrect justification ammonia and HCl is provided</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2     |
| <ul style="list-style-type: none"> <li>• Explains any one of the above concepts well<br/>OR</li> <li>• Outlines both concepts with NO chemical equation/incorrect equation with multiple errors in understanding<br/>OR</li> <li>• No explanation but one correct equation only<br/>OR</li> <li>• Outlines both concepts but no chemical equation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1     |